

AI-Powered Cardiovascular Patient Monitoring System V 1.0

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Table of Contents

Chapter 1 :Introduction	7
1.1 Project Introduction	7
1.2 Existing Examples / Solutions	7
1.3 Business Scope	8
1.4 Useful Tools and Technologies	8
1.5 Project Work Breakdown.....	8
1.6 Project Timeline	9
Chapter 2:Requirement Specification and Analysis	10
2.1 Functional Requirements	10
2.2 Non-Functional Requirements	10
2.3. Selected Functional Requirements	11
2.4. System Use Case Modeling	12
2.5. System Sequence diagram	27
2.6. Domain Model.....	35
Chapter 3: System Design.....	36
3.1. Layer Definition	36
3.2. Software Architecture.....	36
3.3. Class Diagram	38
3.4. Sequence Diagram.....	39

3.5. Entity Relationship Diagram	40
3.6. Database Schema	41
3.7. User Interface Design	43
Chapter 4: Implementation.....	52
4.1. Development Environment.....	52
4.2. System Folder Structure	52
4.3. Backend Implementation (PHP)	52
4.4. Database Implementation (MySQL)	52
4.5. User Interface Implementation.....	53
4.6. Device Integration	57
Chapter 5: Software Testing	58
5.1. Testing Methodology	58
5.2. Test Environment.....	58
5.3. Test Cases	59
5.4. Testing Summary	64

List of Tables

Table 1 1.2.1 Comparison of Existing Cardiovascular Monitoring Systems	7
Table 2 1.2.1 Project timeline	9
Table 3 2.1.1 Functionoal Requirements	10
Table 4 2.2.1 Non Functional requirements	11
Table 5 2.3.1 Selelcted Functional Requirements	11
Table 6 2.4.1 USECASE	16
Table 7 2.4.1.....	16
Table 8 2.4.2.....	17
Table 9 2.4.3.....	18
Table 10 2.4.4.....	19
Table 11 2.4.5.....	20
Table 12 2.4.6.....	21
Table 13 2.4.7.....	22
Table 14 2.4.8.....	23
Table 15 2.4.9.....	24
Table 16 2.4.10.....	25
Table 17 2.4.11.....	26
Table 18 3.1 LAYers Defination	36
Table 20 5.3.1 TC.....	59
Table 21 5.3.2 TC 2.....	60
Table 22 5.3.3 TC 3.....	61
Table 23 5.3. 4 TC 4.....	62
Table 24 5.3.5 TC 5.....	63

List of Figures

Figure 1 1.5.1 project work breakdown	8
Figure 2 2.4.1 Admin Usecase model.....	13
Figure 3 2.4.2 Doctors Usecase model.....	14
Figure 4 2.4.3 Patient usecase model	15
Figure 5 2.5.1 Patient usecaseSSDs.....	27
Figure 6 2.5.2 patient login SSDS	27
Figure 7 2.5.3 Patient vital sign view	28
Figure 8 2.5.4 Patient medical history view.....	28
Figure 9 2.5.5 patient preious view history	28
Figure 10 2.5.6 Doctors Registration SSD	29
Figure 11 2.5.7 Doctors login SSDs.....	29
Figure 12 2.5.8 Doctor add patients SSD	30
Figure 13 2.5.9 Doctor edits patient SSDs.....	30
Figure 14 2.5.10 Doctor show previous patients history.....	31
Figure 15 2.5.11 Doctor view list of patients	31
Figure 16 2.5.12 Admin login	31
Figure 17 2.5.13 Admin add patients.....	32
Figure 18 2.5.14 Admin add devices	32
Figure 19 2.5.15 admin add doctor event schedule	32
Figure 20 2.5.16 Admin can view doctor list.....	33
Figure 21 2.5.17 Admin patient list and count.....	33
Figure 22 2.5.18 admin view all devices list.....	34
Figure 23 2.6.1 Domain Model	35
Figure 24 3.2.1 Software Architecture.....	37
Figure 25 3.3.1 Class Diagram	38
Figure 26 3.4.1 Sequence Diagram	39
Figure 27 3.5.1 ERD	40
Figure 28 3.6.1 Database Schema.....	42
Figure 29 3.7.1 login page	43
Figure 30 3.7.2 signup page	43
Figure 31 3.7.3 signup page	44
Figure 32 3.7.4 Admins Dashboard.....	44
Figure 33 3.7.5 Admin Dashboard overview.....	45
Figure 34 3.7.6 Add doctors	45
Figure 35 3.7.7 Doctors Schedule	46
Figure 36 3.7.8 Add events	46
Figure 37 3.7.9 Patients view	47

Figure 38 3.7.10 Edit Patient	47
Figure 39 3.7.11 Patients Profile	48
Figure 40 3.7.12 add patients	48
Figure 41 3.7.13 ECG Devices.....	49
Figure 42 3.7.14 Add devices	49
Figure 43 3.7.15 Doctors Dashboard	50
Figure 44 3.7.16 Doctors edit patients.....	50
Figure 45 3.7.17 Doctor can add patients.....	51
Figure 46 3.7. 18 Patients Dashboard	51
Figure 47 4.5.1 Admin login page	53
Figure 48 4.5.2 admin dashboard	54
Figure 49 4.5.3 Add doctor page.....	54
Figure 50 4.5.4 view doctors	55
Figure 51 4.5.5 add device page	55
Figure 52 4.5.6 medical history.....	56
Figure 53 4.5.7 API Modules	57

Chapter 1 :Introduction

1.1 Project Introduction

This project presents a web-based Cardiovascular Patient Monitoring System designed to manage patient information, visualize physiological data, and support remote health monitoring through a centralized dashboard. The system is implemented using PHP and MySQL and communicates with external devices using REST-based APIs. Raspberry Pi devices are supported as monitoring units for data transmission.

The current implementation focuses on user management, patient records, device management, and dashboard-based visualization of vital data. Advanced features such as medical-grade sensor integration (ADS1292R) and AI-based disease prediction are planned for future phases.

1.2 Existing Examples / Solutions

Existing cardiovascular monitoring solutions such as Holter monitors, Apple Watch ECG, and mobile ECG devices provide basic heart monitoring capabilities. However, these systems are often expensive, closed-source, or lack integration with hospital information systems. The proposed system aims to provide a flexible, low-cost, and extensible academic prototype.

Table 1 1.2.1 Comparison of Existing Cardiovascular Monitoring Systems

Sr. No.	System	AI Integration	Real-Time Monitoring	Cost Efficiency	Hospital Integration
1	Holter Monitor	X	✓	X	✓
2	Apple Watch ECG	✓	✓	X	X
3	Mobile	✓	✓	X	X
4	Proposed System	✓	✓	✓	✓

1.3 Business Scope

The system can be used in hospitals, clinics, telemedicine platforms, and home-based monitoring. It supports digital health record management and can be extended for remote patient monitoring and future AI-assisted decision support.

1.4 Useful Tools and Technologies

Proposed Technology Stack

The current implementation uses PHP for backend development, MySQL for database management, and HTML, CSS, and JavaScript for frontend interfaces. REST APIs are used for data communication. Raspberry Pi devices act as monitoring gateways. AI frameworks and advanced sensors are considered future enhancements.

1.5 Project Work Breakdown

The project is divided into requirement analysis, system design, software development, testing, and documentation. Hardware upgrades and AI integration are planned in later stages. A sample work breakdown structure is illustrated below.

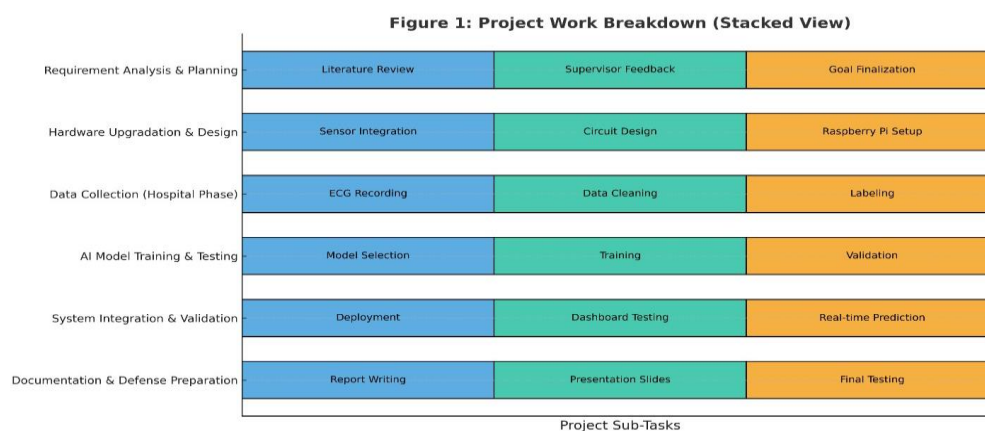


Figure 1 1.5.1 project work breakdown

1.6 Project Timeline

The following table represents the estimated project timeline, showing the distribution of tasks across weeks.

Table 2 1.2.1 Project timeline

Phase	Duration	Key Activities
Requirement Analysis & Planning	Oct 2025	Supervisor feedback, literature review, planning hospital visits
Hardware Upgradation & Design	Nov – Dec 2025	Purchase & integrate new sensors, test signal quality
Data Collection (Hospital Phase)	Jan – Feb 2026	Collect ECG readings from 60–70 patients at RIC
Data Labeling & Preprocessing	Mar 2026	Label and clean collected ECG data
AI Model Training & Testing	Apr – May 2026	Train models, evaluate performance, tune parameters
System Integration & Validation	Jun 2026	Deploy model into system, test real-time predictions
Final Documentation & Defense Preparation	Jul 2026	Report writing, final testing, and FYP defense prep

Chapter 2: Requirement Specification and Analysis

Requirements analysis is the process of determining and documenting the expectations for the proposed system's improvements. Since the project is upgrading to the medical-grade **ADS1292R** and implementing a multi-class AI, these requirements are rigorous, focusing on clean data acquisition and accurate disease classifications.

2.1 Functional Requirements

Functional Requirements are the specific actions the system must be able to perform.

Table 3 2.1.1 Functional Requirements

S No	Functional Requirement	Type	Status
1	Secure User Registration & Authentication	Core	Pending
2	Patient Profile & Medical History Management	Core	Pending
3	Role-Based Access Control (RBAC)	Core	Pending
4	Real-Time Dashboard Visualization (ECG, HR, SpO ₂ , Respiration)	Core	Pending
5	Device Registration & Management (Raspberry Pi)	Core	Pending
6	Upgraded Sensor Data Acquisition (ADS1292R – 24-bit ECG/Respiration)	Core	Pending
7	Real-Time Data Pre-processing using ADS1292R	Core	Pending
8	Multi-Class AI Prediction Service	Intermediate	Pending
9	Critical Alert Generation (Edge)	Core	Pending
10	AI Prediction History & Auditing	Intermediate	Pending
11	Data Export (EHR Compatibility)	Intermediate	Planned

2.2 Non-Functional Requirements

Non-Functional Requirements are about the **quality** of the system—how reliable, fast, and secure it is.

Table 4 2.2.1 Non Functional requirements

S No	Non-Functional Requirements	Category
1	The system shall enforce secure authentication and role-based access control to protect patient data.	Security
2	Secure communication between users and the web system shall be supported using standard web security practices (HTTPS depending on deployment).	Security
	The system dashboard shall respond within an acceptable time under normal operating conditions.	Performance
3	The system shall be reliable for continuous monitoring during normal usage.	Reliability
4	The system architecture shall support future scalability for AI integration and additional devices.	Scalability
5	The system shall be usable on standard web browsers with a simple and intuitive interface.	Usability
6	Advanced AI accuracy metrics and real-time latency constraints are defined as target goals for future phases.	AI Performance (Planned)
7	The system shall enforce secure authentication and role-based access control to protect patient data.	Security

2.3. Selected Functional Requirements

Following is the list of the requirements selected for the current iteration.

Table 5 2.3.1 Selected Functional Requirements

S. No.	Functional Requirement	Type
1	Upgraded Sensor Data Acquisition using ADS1292R (24-bit ECG, Respiration)	Core
2	Real-Time Signal Pre-processing on Raspberry Pi	Core
3	Multi-Class AI Prediction Service	Core

4	Critical Alert Generation at Edge Device	Core
5	Enhanced Real-Time Dashboard Visualization (ECG, HR, SpO ₂ , Respiration)	Intermediate
6	Patient Medical History with AI Prediction Logs	Intermediate

NOTE :

“Only User Management, Patient Records, Device Management, And Dashboard Visualization Are Implemented In The Current Phase. The Following Requirements Represent The Planned Implementation Roadmap.”

2.4. System Use Case Modeling

The system includes Admin, Doctor, and Patient actors. Use cases focus on authentication, patient management, device management, and data visualization.

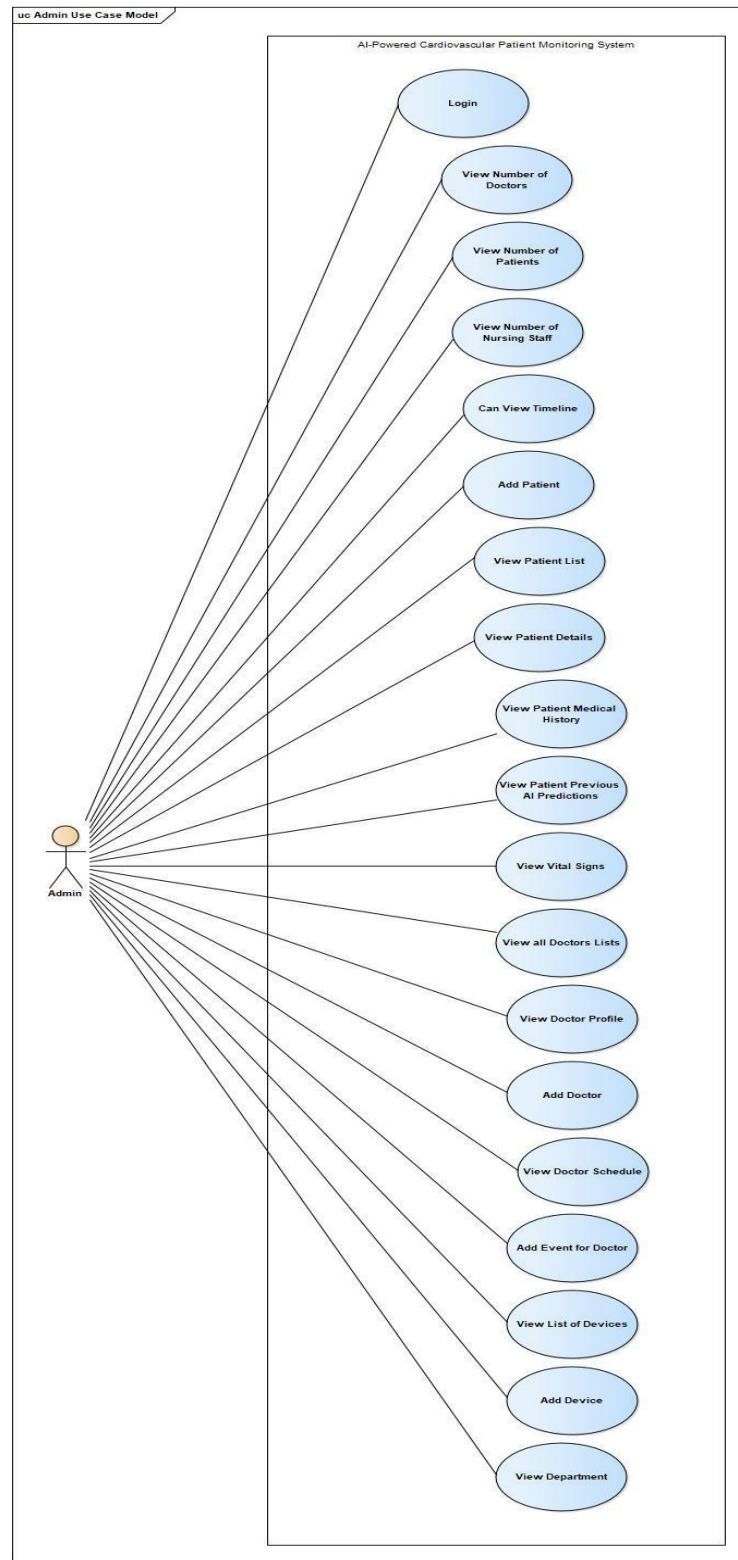


Figure 2 2.4.1 Admin Usecase model



Figure 3 2.4.2 Doctors Usecase model

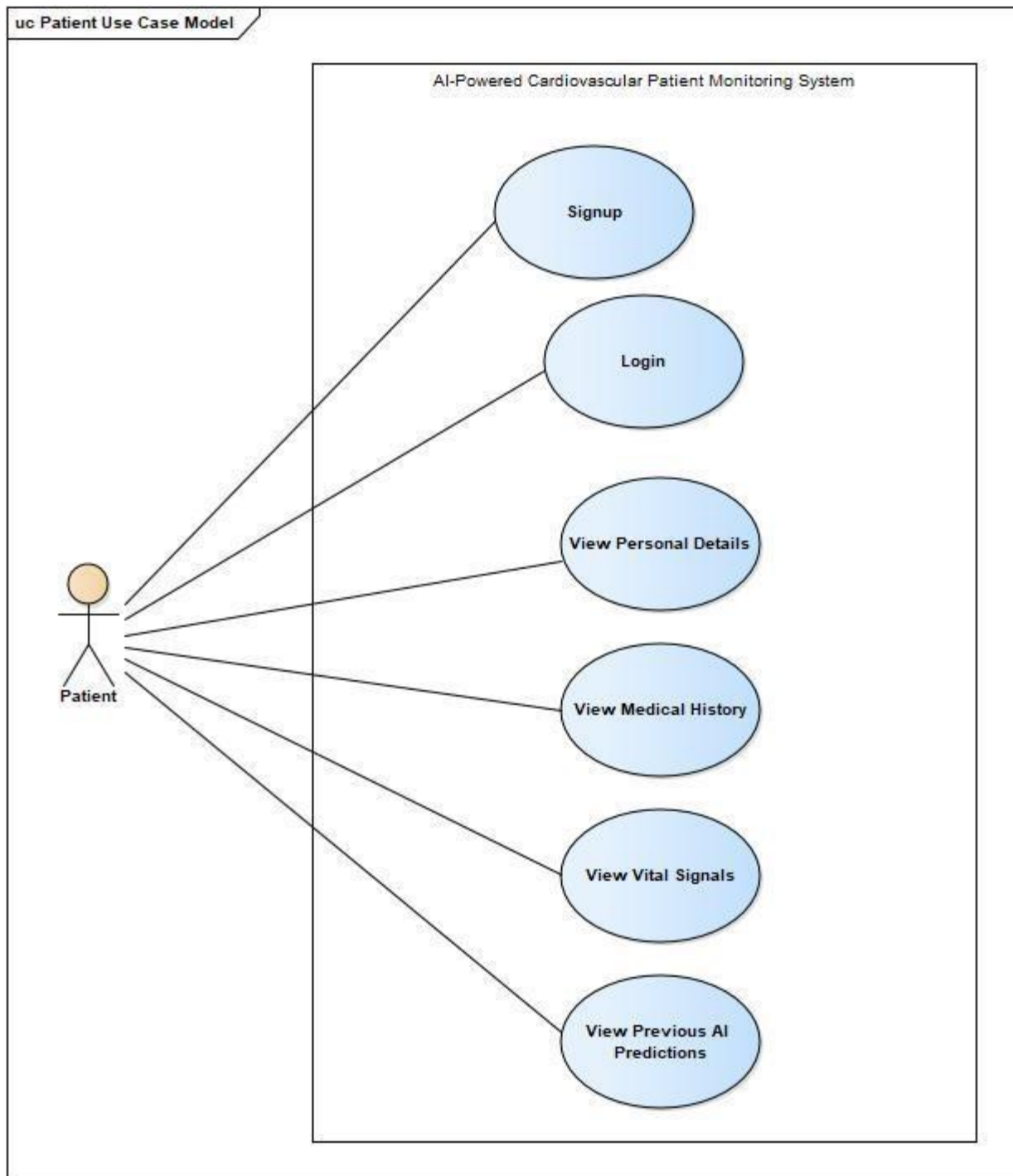


Figure 4 2.4.3 Patient usecase model

Table 6 2.4.1 USECASE

Table 7 2.4.1

Use Case ID:	Uc1		
Use Case Name:	User Signup.		
Created By:	Ehsan Hayat	Last Updated By:	Ehsan Hayat
Date Created:	06-11-25	Last Revision Date:	06-11-25
Actors:	Patient		
Description:	The Patient creates a new account to access personal health data and alerts.		
Trigger:	Patient clicks sign up.		
Preconditions:	Patient has no account.		
Postconditions:	Account created.		
Normal Flow:	Patient	System	
	Initiates action.	Displays signup form.	
	Enters name, mobile number, and password.	Validates input format.	
	Submits data.	Encrypts password, stores data, and responds with confirmation.	
Alternative Flows:	The Patient cancels the operation. / The System auto-populates address based on location.		
Exceptions:	Invalid input format (e.g., weak password). / Database connection error.		

Table 8 2.4.2

Use Case ID:	Uc2		
Use Case Name:	Login		
Created By:	Ehsan Hayat	Last Updated By:	Ehsan Hayat
Date Created:	06-11-25	Last Revision Date:	06-11-25
Actors:	Patient		
Description:	The Patient securely accesses the system using valid credentials.		
Trigger:	The Patient clicks the 'Login' button.		
Preconditions:	The Patient has an existing, active account.		
Postconditions:	The Patient is authenticated and redirected to the personal dashboard.		
Normal Flow:	Patient	System	
	Initiates action.	Displays login form.	
	Enters username and password.	Validates credentials against stored data.	
	Submits data.	Generates access token and redirects to the dashboard.	
Alternative Flows:	The Patient uses 'Forgot Password' link.		
Exceptions:	Invalid username or password. / Server timeout during authentication.		

Table 9 2.4.3

Use Case ID:	Uc3		
Use Case Name:	View Vital Signals		
Created By:	Ehsan Hayat	Last Updated By:	Ehsan Hayat
Date Created:	06-11-25	Last Revision Date:	06-11-25
Actors:	Patient		
Description:	The Patient views the real-time stream of physiological data (ECG, Respiration, SpO2) collected by their personal device.		
Trigger:	The Patient selects the 'View Vital Signals' option on the dashboard.		
Preconditions:	The Patient is logged in; the Monitoring Device is online and streaming data.		
Postconditions:	The system displays a live graphical view of the Patient's vital signs.		
Normal Flow:	Patient	System	
	Requests to view vital signs.	Authenticates the request and fetches the live stream from the Edge Gateway.	
	Views live ECG waveform and numerical readings (HR, SpO2, Respiration Rate).	Displays data and continuously updates the graphs (FR-6.0).	
Alternative Flows:	The Patient filters the view to show only the last 24 hours of readings.		
Exceptions:	The Monitoring Device goes offline (System displays 'Offline'). / No historical data is found.		

Table 10 2.4.4

Use Case ID:	Uc4		
Use Case Name:	View previous AI predictions		
Created By:	Ehsan Hayat	Last Updated By:	Ehsan Hayat
Date Created:	06-11-25	Last Revision Date:	06-11-25
Actors:	Patient		
Description:	The Patient reviews the historical records of the system's automated multi-class disease predictions.		
Trigger:	The Patient selects the 'View Previous AI Predictions' option.		
Preconditions:	The Patient is logged in; the AI prediction log exists for the patient.		
Postconditions:	The system displays a chronological log of all past AI diagnoses, scores, and timestamps.		
Normal Flow:	Patient	System	
	Requests the prediction history.	Retrieves the AIPredictionLog from the database (FR-8.0).	
	Views the list of prediction classes (e.g., 'Normal', 'Tachycardia') and confidence scores.	Displays the log, highlighting any 'Critical Alerts' (FR-5.0).	
Alternative Flows:	The Patient filters the log by date range or prediction class.		
Exceptions:	No AI predictions are recorded. / Database query error.		

Table 11 2.4.5

Use Case ID:	Uc5		
Use Case Name:	View Medical History		
Created By:	Ehsan Hayat	Last Updated By:	Ehsan Hayat
Date Created:	06-11-25	Last Revision Date:	06-11-25
Actors:	Patient		
Description:	The Patient views non-real-time medical summaries and associated doctor information.		
Trigger:	The Patient selects the 'View Medical History' tab.		
Preconditions:	The Patient is logged in.		
Postconditions:	The system displays patient demographic information and associated Doctor details.		
Normal Flow:	Patient	System	
	Requests medical history.	Fetches static medical records and assigned Doctor details (FR-7.0).	
	Reviews summary and Doctor's contact information.	Displays the fetched information securely.	
Alternative Flows:	The Patient prints or exports the record summary (FR-10.0).		
Exceptions:	Database retrieval fails. / No medical history is recorded.		

Table 12 2.4.6

Use Case ID:	Uc6		
Use Case Name:	Add Patient		
Created By:	Ehsan Hayat	Last Updated By:	Ehsan Hayat
Date Created:	06-11-25	Last Revision Date:	06-11-25
Actors:	Doctor		
Description:	The Doctor registers a new patient into the monitoring system and assigns them to an available device.		
Trigger:	The Doctor clicks the 'Add Patient' button.		
Preconditions:	The Doctor is logged in; the Doctor has permission to add patients.		
Postconditions:	A new patient record is created and linked to the Doctor's profile.		
Normal Flow:	Doctor	System	
	Requests to add patient.	Displays patient creation form.	
	Enters patient details (name, date of birth, contact, etc.).	Validates data.	
	Submits details.	Creates patient entry and confirms registration.	
Alternative Flows:	The Doctor attempts to assign the patient to a device already in use. $\rightarrow$ System warns the Doctor and requests a different device ID.		
Exceptions:	Invalid or missing mandatory fields. / Database write error.		

Use Case ID:	Uc7		
Use Case Name:	Track All Previous AI Predictions		
Created By:	Ehsan Hayat	Last Updated By:	Ehsan Hayat
Date Created:	06-11-25	Last Revision Date:	06-11-25
Actors:	Doctor		
Description:	The Doctor reviews the comprehensive historical AI diagnosis log for any assigned patient, including prediction classes and confidence scores.		
Trigger:	The Doctor selects a patient and clicks 'Track All previous AI Predictions'		
Preconditions:	The Doctor is logged in and authorized for the patient; prediction data exists.		
Postconditions:	A sortable, filterable table of the patient's full prediction history is displayed (FR-8.0).		
Normal Flow:	Doctor	System	
	Selects the patient.	Verifies authorization (FR9.0).	
	Requests full prediction log.	Retrieves all AIPredictionLog records for the Patient ID.	
	Filters the view to show only high-confidence abnormal readings.	Displays filtered data in a sortable log format.	
Alternative Flows:	The Doctor exports the prediction history log (FR-10.0).		
Exceptions:	The Doctor is not authorized to view the patient. / Database access fails.		

Table 13 2.4.7

Use Case ID:	Uc8		
Use Case Name:	Can Look Vital Signs (Real-Time Monitoring)		
Created By:	Ehsan Hayat	Last Updated By:	Ehsan Hayat
Date Created:	06-11-25	Last Revision Date:	06-11-25
Actors:	Doctor		
Description:	The Doctor views the real-time multi-parameter data streams (ECG, Respiration, SpO2) of an active patient for immediate clinical assessment.		
Trigger:	The Doctor selects a patient from the 'View List of Patients' dashboard.		
Preconditions:	The Doctor is logged in; the Patient's Monitoring Device is actively streaming data (FR-2.0).		
Postconditions:	A live dashboard showing all vital sign graphs and the latest AI prediction is displayed (FR-6.0).		
Normal Flow:	Doctor	System	
	Selects patient for real-time viewing.	Fetches current status and opens the monitoring dashboard.	
	Monitors live ECG waveform, Respiration Rate, and SpO2 trends.	Displays dynamic graphs and instantaneous AI prediction updates (NFR-3.0).	
	Observes a critical alert flash.	Logs alert acknowledgement when the Doctor clicks on it (FR-5.0).	
Alternative Flows:	The Doctor pauses the live data stream to inspect a segment of the ECG waveform more closely.		
Exceptions:	The live data stream connection fails. / The Doctor is unauthorized to view the patient.		

Table 14 2.4.8

Table 15 2.4.9

Use Case ID:	Uc9		
Use Case Name:	Add Doctor / Add Event for Doctor		
Created By:	Ehsan Hayat	Last Updated By:	Ehsan Hayat
Date Created:	06-11-25	Last Revision Date:	06-11-25
Actors:	Admin		
Description:	The Admin manages the healthcare workforce by adding new Doctor accounts or scheduling new events/appointments for them.		
Trigger:	The Admin selects 'Add Doctor' or 'Add Event for Doctor' from the management menu.		
Preconditions:	The Admin is logged in and authorized (FR-9.0).		
Postconditions:	A new Doctor account is created, or an event/schedule entry is added to the Doctor's record.		
Normal Flow:	Admin	System	
	Requests to add Doctor or add event.	Displays the corresponding input form.	
	Enters required credentials (for new doctor) or schedule details (for event).	Validates inputs and updates the User and Doctor tables.	
	Submits data.	Confirms successful creation/update.	
Alternative Flows:	The Admin modifies an existing Doctor's details or schedule.		
Exceptions:	A Doctor with that ID already exists. / Invalid date/time format for the event.		

Table 2.10: view Doctor Profile Usecase Description

Use Case ID:	Uc10		
Use Case Name:	View Doctor Profile / View all Doctors Lists		
Created By:	Ehsan Hayat	Last Updated By:	Ehsan Hayat
Date Created:	06-11-25	Last Revision Date:	06-11-25
Actors:	Admin		
Description:	The Admin views a list of all registered doctors and can drill down to view individual Doctor profiles and schedules.		
Trigger:	The Admin selects 'View all Doctors Lists'.		
Preconditions:	The Admin is logged in.		
Postconditions:	The list of Doctors or a specific Doctor's profile is displayed.		
Normal Flow:	Admin	System	
	Requests the Doctor list.	Retrieves all Doctor records.	
	Views the list and selects a specific Doctor.	Displays the selected Doctor's profile, including schedule and managed patients.	
Alternative Flows:	The Admin views records for other staff ('View Number of Nursing Staff').		
Exceptions:	Database retrieval fails. / No Doctors are currently registered.		

Table 2.11: Add Device Usecase Description

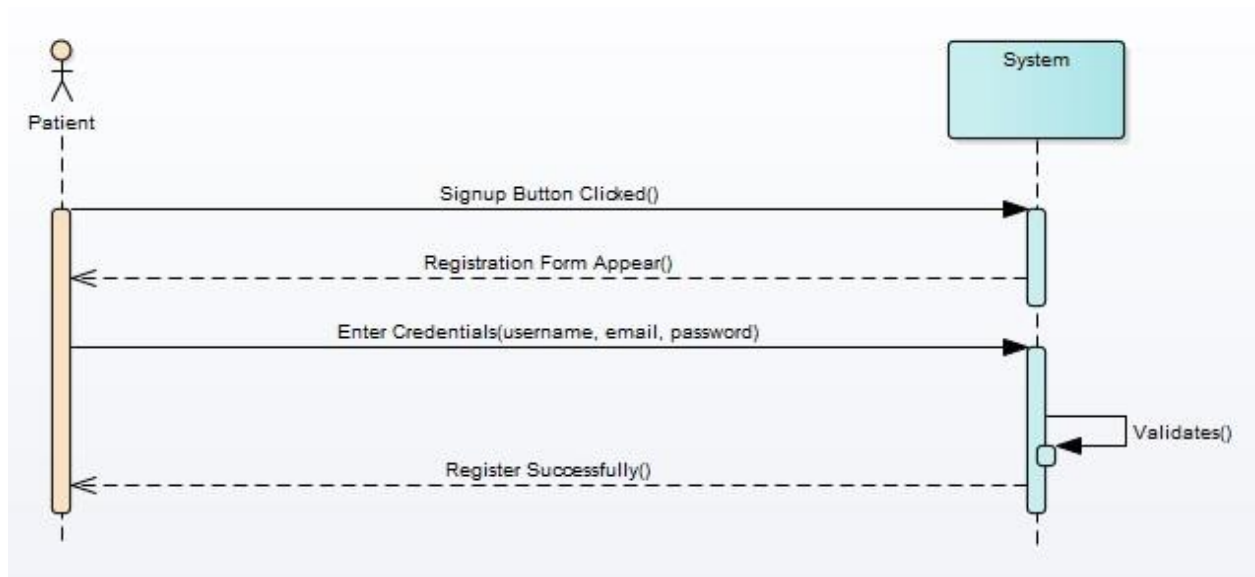
Use Case ID:	Uc11		
Use Case Name:	Add Device / View List of Devices		
Created By:	Ehsan Hayat	Last Updated By:	Ehsan Hayat
Date Created:	06-11-25	Last Revision Date:	06-11-25
Actors:	Admin		
Description:	The Admin registers new Monitoring Devices (Raspberry Pi units) into the system inventory and manages their status.		
Trigger:	The Admin selects 'Add Device' or 'View List of Devices'.		
Preconditions:	The Admin is logged in.		
Postconditions:	A new device is registered, or the inventory list is displayed.		
Normal Flow:	Admin	System	
	Requests to add device.	Displays device registration form.	
	Enters device serial number and attributes.	Validates and stores the new device record.	
	Views the updated device list.	Retrieves and displays all device records, including online/offline status.	
Alternative Flows:	The Admin modifies the status or assigned patient of an existing device.		
Exceptions:	Device serial number already exists. / Database write error.		

2.5. System Sequence diagram

Sequence diagrams illustrate interactions for login, patient registration, device assignment, and dashboard access.

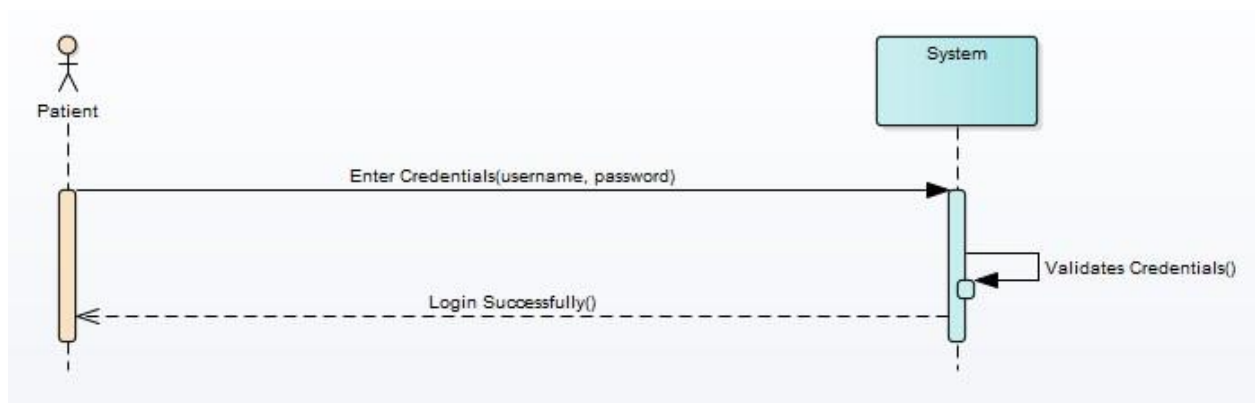
Patient Account Registration

Figure 5 2.5.1 Patient usecaseSSDs



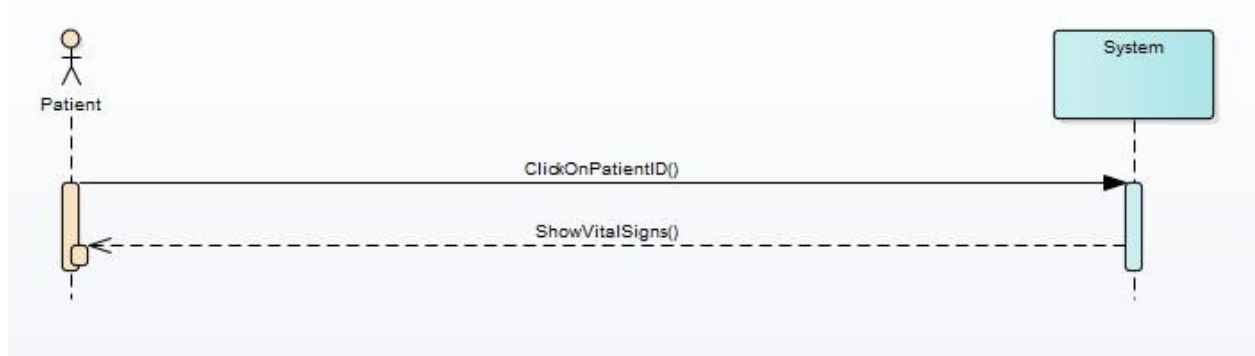
Patient System Login

Figure 6 2.5.2 patient login SSDS



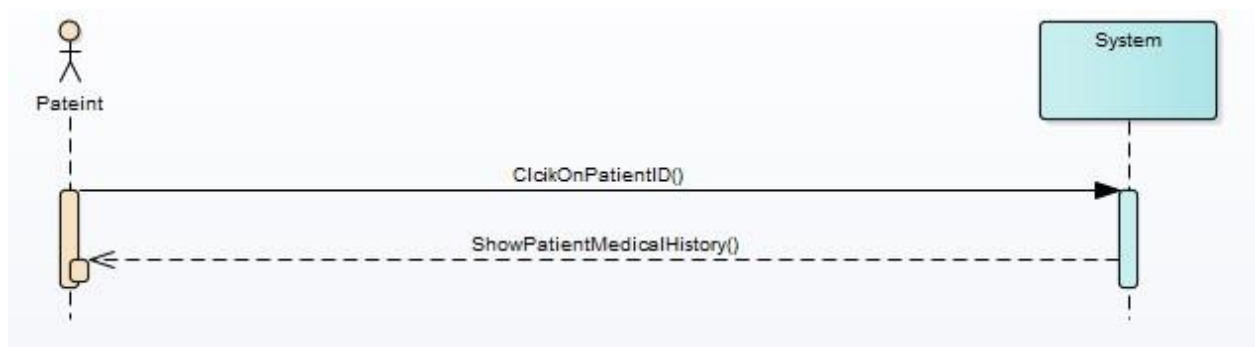
Patient View Vital Signs

Figure 7 2.5.3 Patient vital sign view



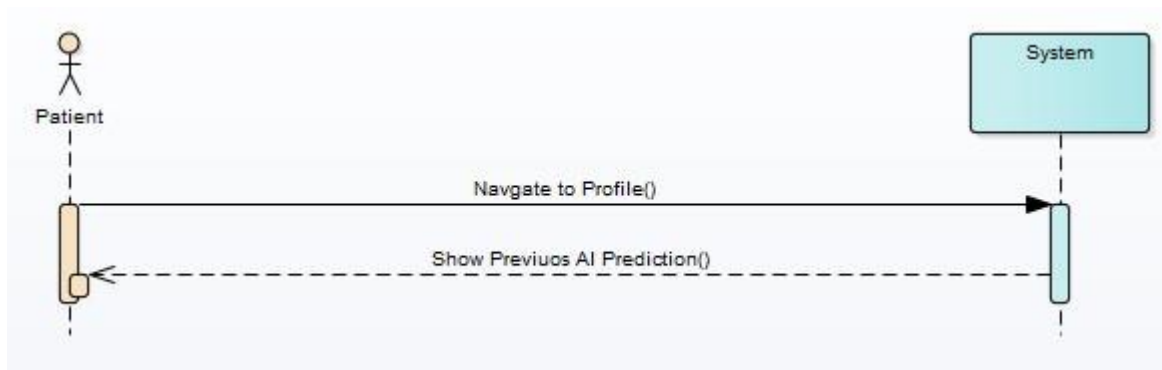
Patient View Medical History

Figure 8 2.5.4 Patient medical history view



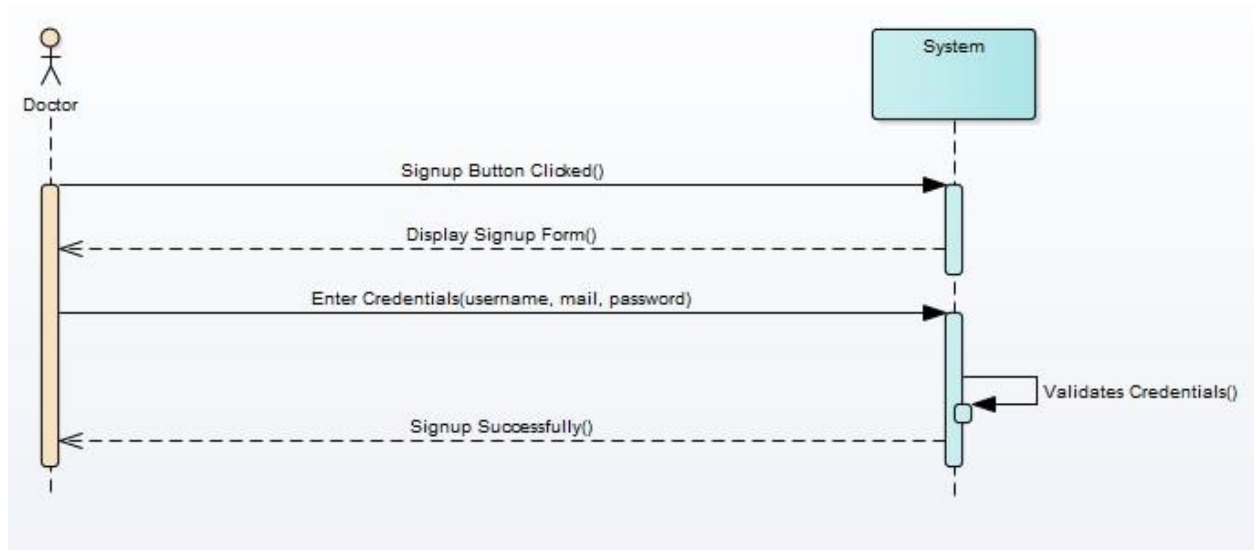
Patient View Previous AI Predictions

Figure 9 2.5.5 patient previous view history



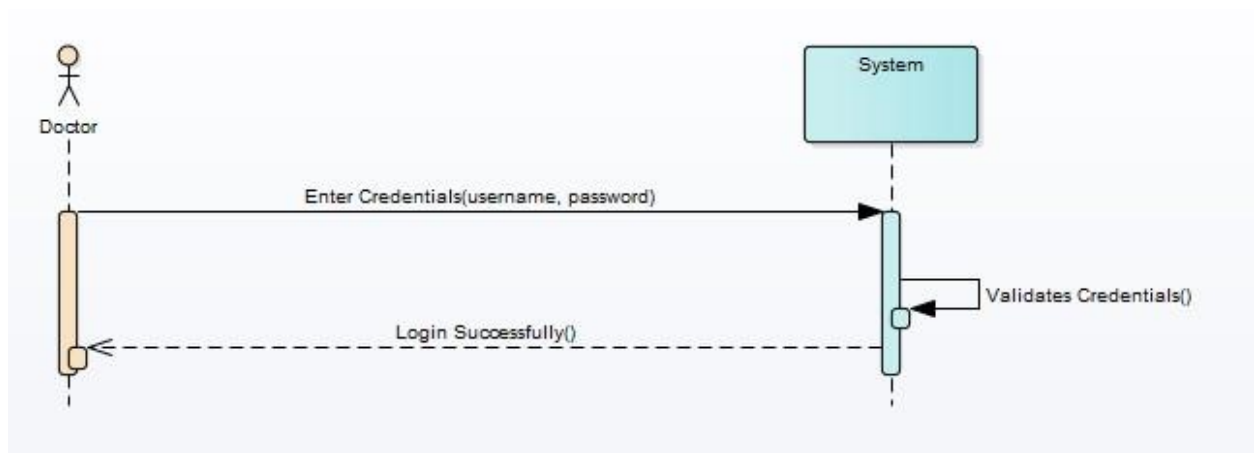
Doctor Account Registration

Figure 10 2.5.6 Doctors Registration SSD



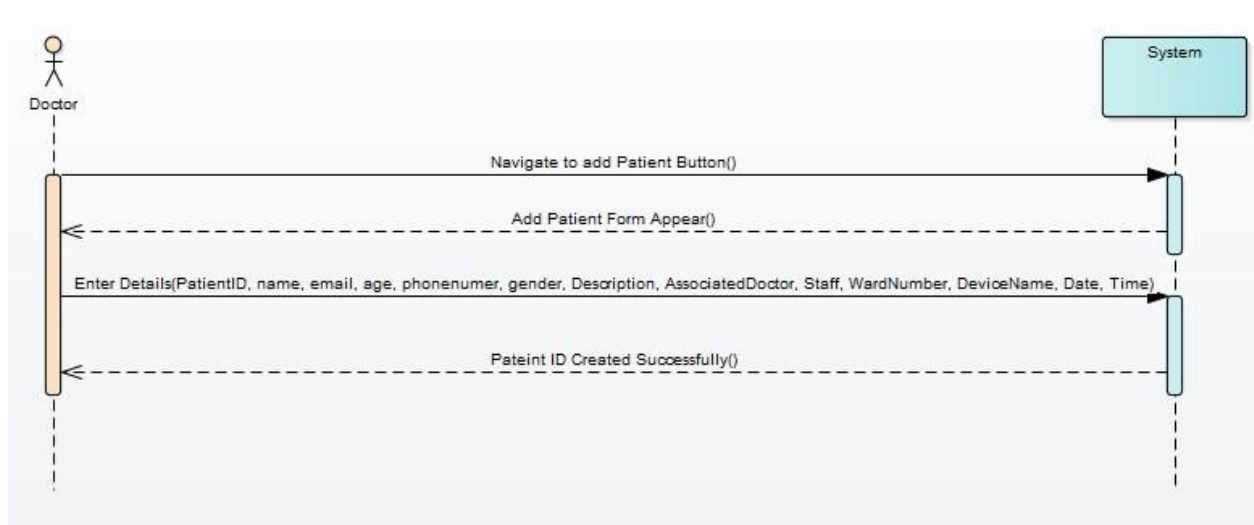
Doctor System Login

Figure 11 2.5.7 Doctors login SSDs



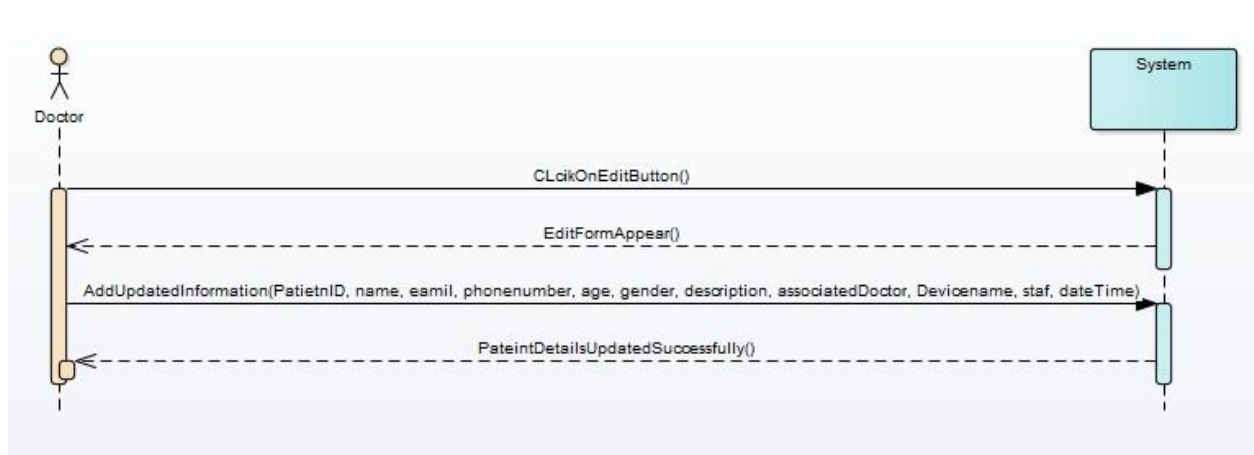
Doctor Add Patient

Figure 12 2.5.8 Doctor add patients SSD



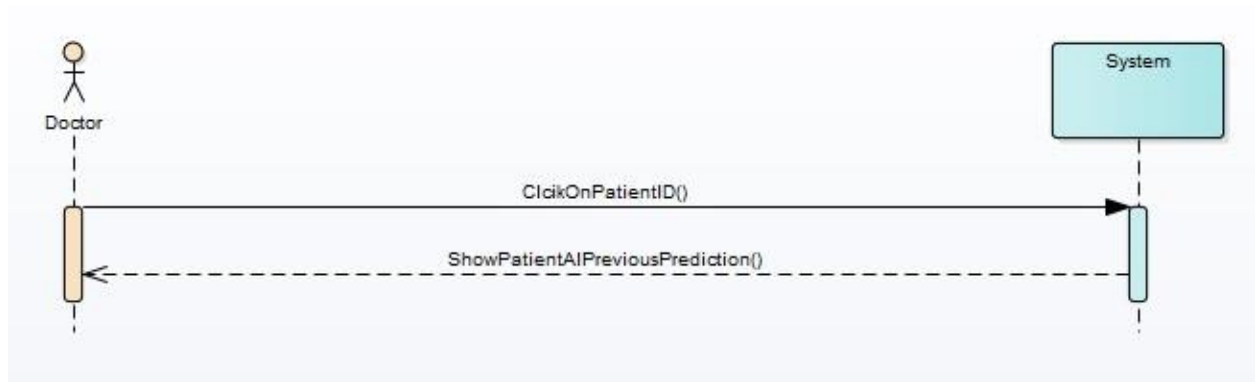
Doctor Edit Patient Data

Figure 13 2.5.9 Doctor edits patient SSDs



Doctor Track All AI Predictions

Figure 14 2.5.10 Doctor show previous patients history



Doctor View List of Patients

Figure 15 2.5.11 Doctor view list of patients



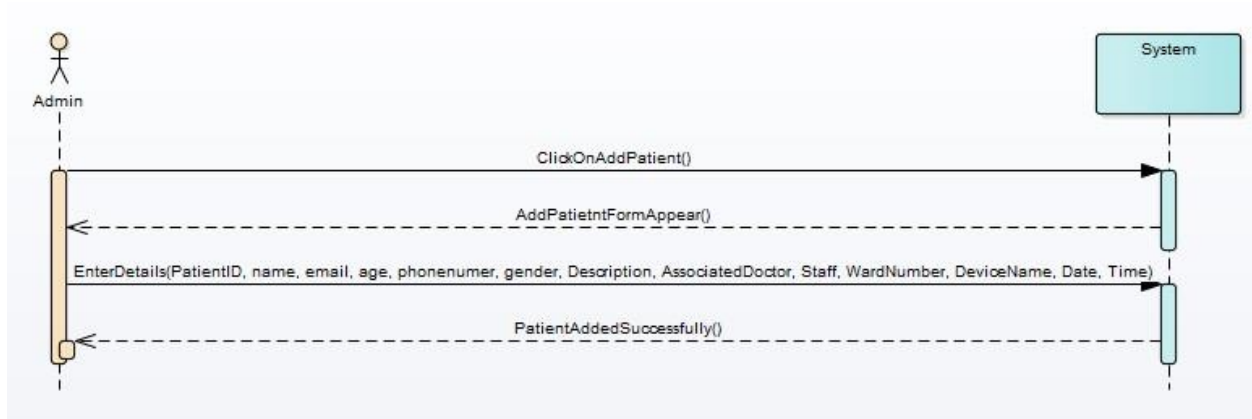
Admin System Login

Figure 16 2.5.12 Admin login



Admin Add Patient

Figure 17 2.5.13 Admin add patients



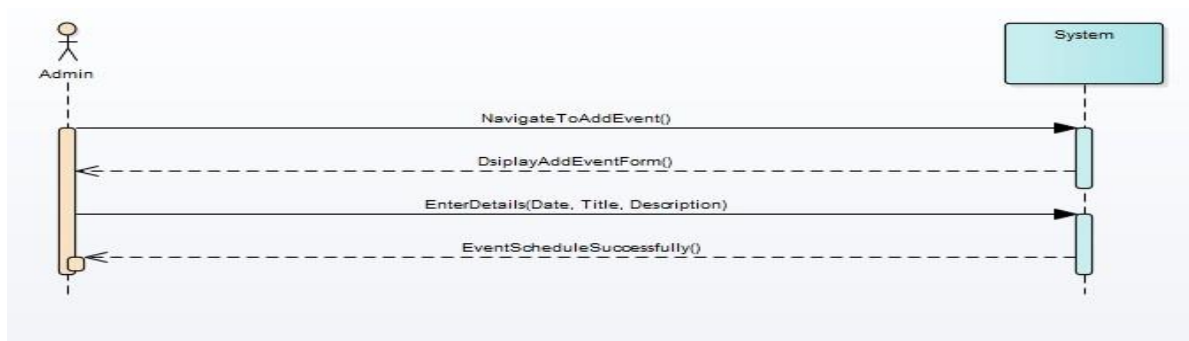
Admin Add Device

Figure 18 2.5.14 Admin add devices



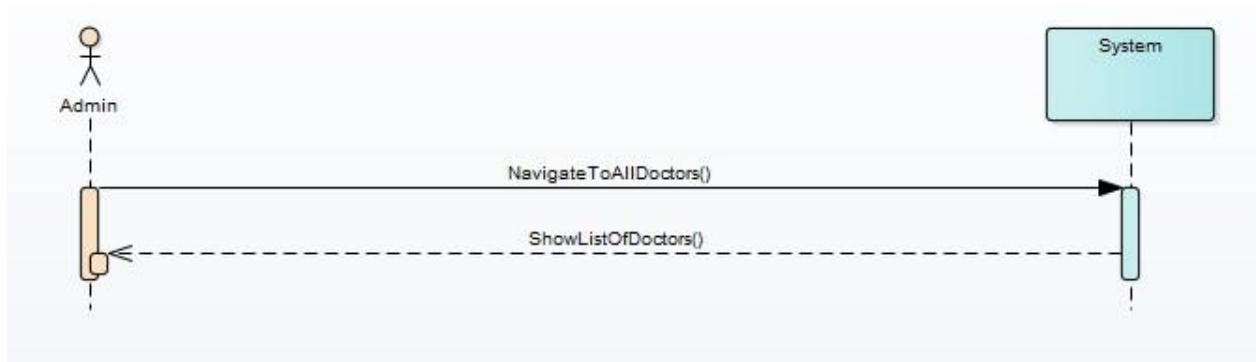
Admin Add Doctor Event/Schedule

Figure 19 2.5.15 admin add doctor event schedule



Admin View All Doctors List

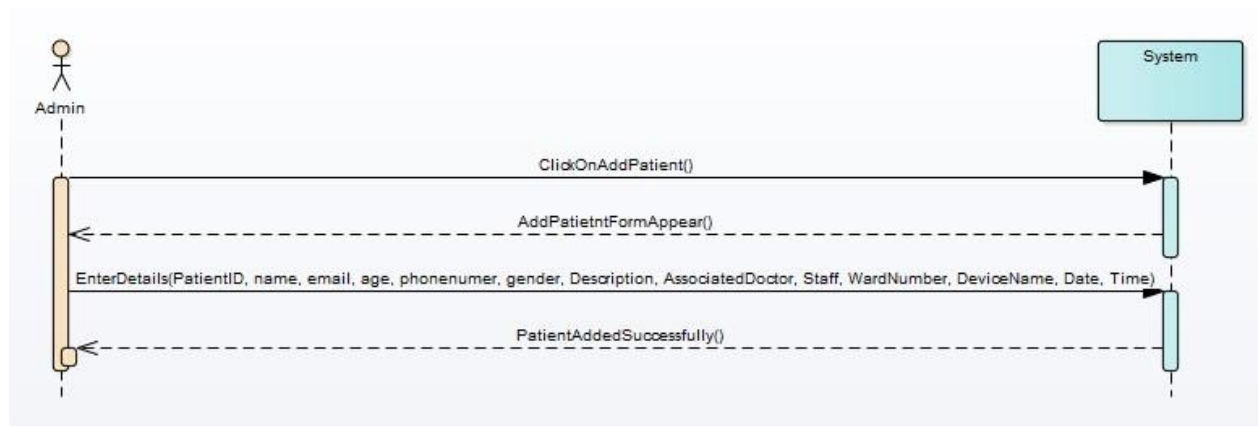
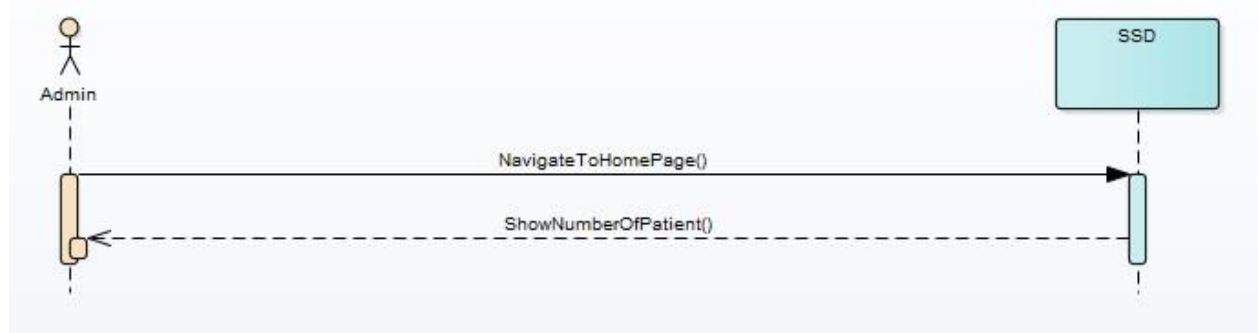
Figure 20 2.5.16 Admin can view doctor list



Admin View Patient List & Count

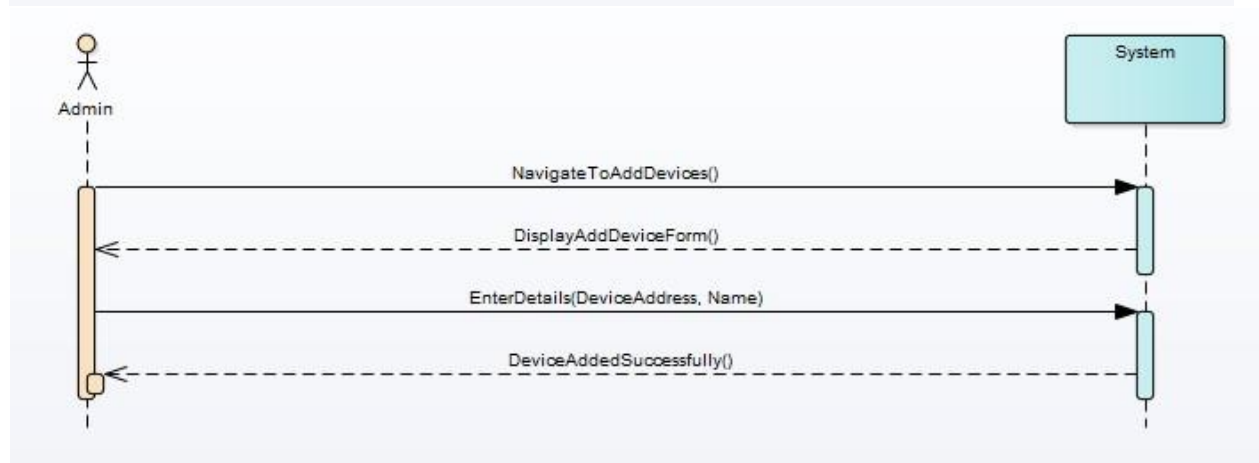
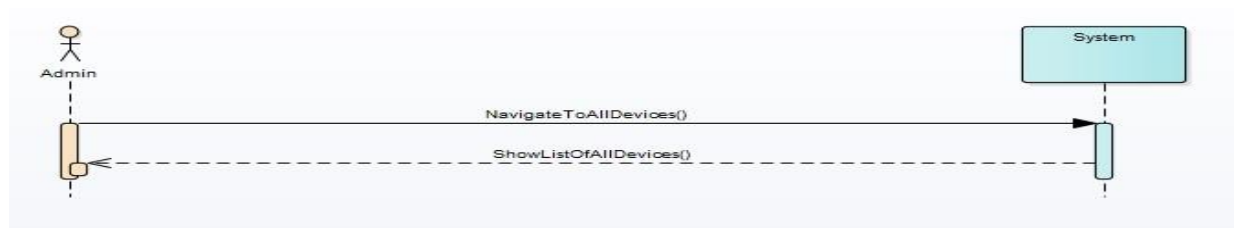
Figure 21 2.5.17 Admin patient list and count





Admin View All Devices List

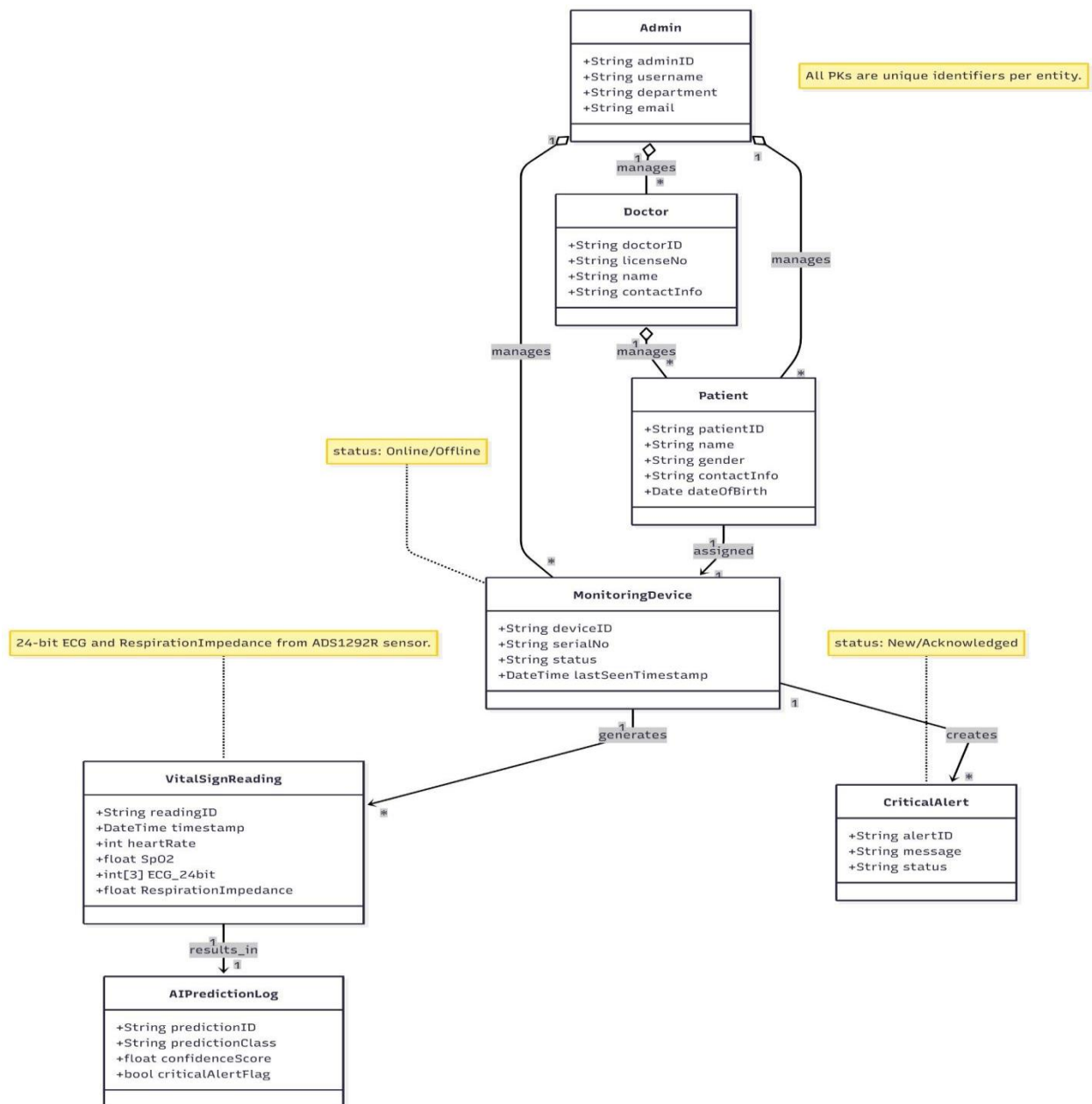
Figure 22 2.5.18 admin view all devices list



2.6. Domain Model

The domain model defines core entities such as User, Patient, Doctor, Device, and Vital Data. AI-related entities are reserved for future integration.

Figure 23 2.6.1 Domain Model



Chapter 3: System Design

3.1. Layer Definition

The system follows a three-tier architecture consisting of Presentation Layer, Business Logic Layer, and Database Layer.

Table 18 3.1 Layers Defination

Layers	Description
Presentation Layer	This layer will be used for the interaction with the user through a graphical user interface (The React.js Dashboard) to display real-time data (FR-6.0).
Business Logic Layer	This layer contains the core application logic (Django Backend). All the constraints, user authentication (FR-9.0), and data routing functions reside under this layer.
Database Layer	This layer contains the database servers (PostgreSQL/MongoDB) where all permanent patient information and historical logs (FR-8.0) are stored and retrieved securely.

3.1.1. Presentation Layer

This tier is the top level that displays information and interacts with the user via a responsive graphical interface.

3.1.2. Business Logic Layer

Also called the middle tier, this layer controls application functionality by performing detailed processing, enforcing security rules, and managing communication between the database and the presentation layer.

3.1.3. Database Layer

This tier includes database servers where information is stored and retrieved. Data in this tier is kept independent of application servers, ensuring robust data integrity and security.

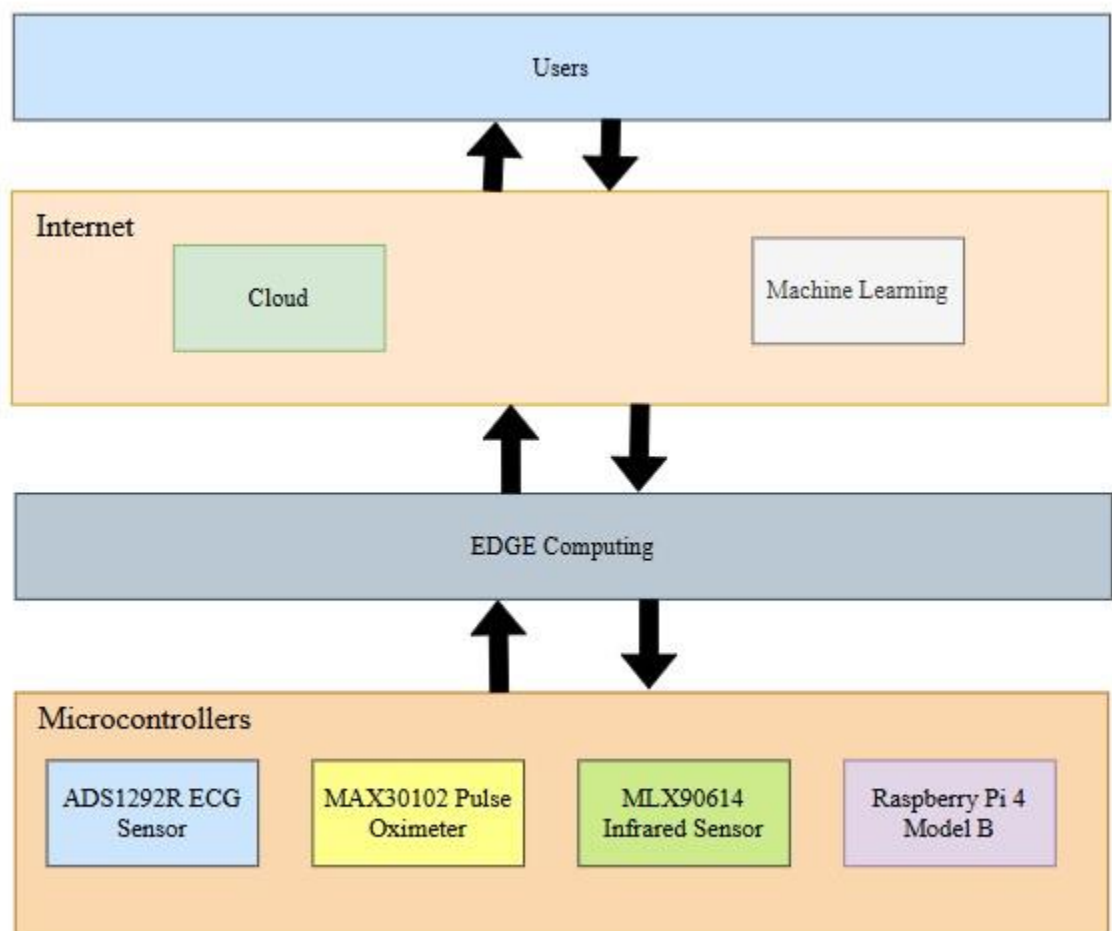
3.2. Software Architecture

The architecture is a web-based system integrated with IoT data sources via APIs. PHP handles backend logic, while MySQL stores persistent data. AI processing is planned as a future enhancement.

- **Architecture:** Hybrid Edge-Cloud Architecture.

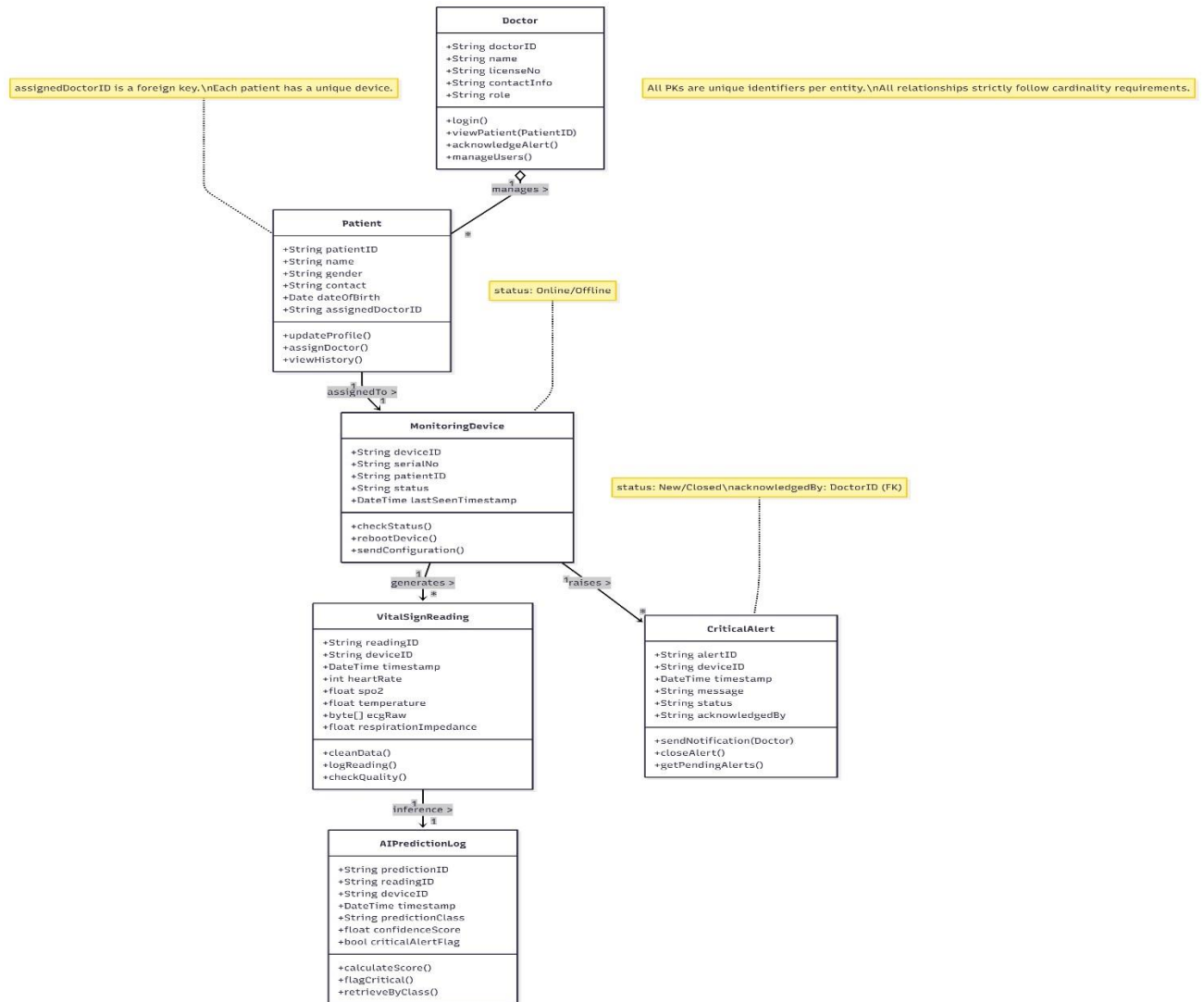
- **Sensing Network:** The **ADS1292R (24-bit ECG/Respiration)** and **MAX30102 (\$SpO_{2}\$)** acquire the data.
- **Edge Processing Node (Raspberry Pi 4):** Performs local pre-processing and AI Inference (FR-3.0, FR-4.0).
- **Cloud API Gateway:** Secure entry point for all data and user requests.

Figure 24 3.2.1 Software Architecture



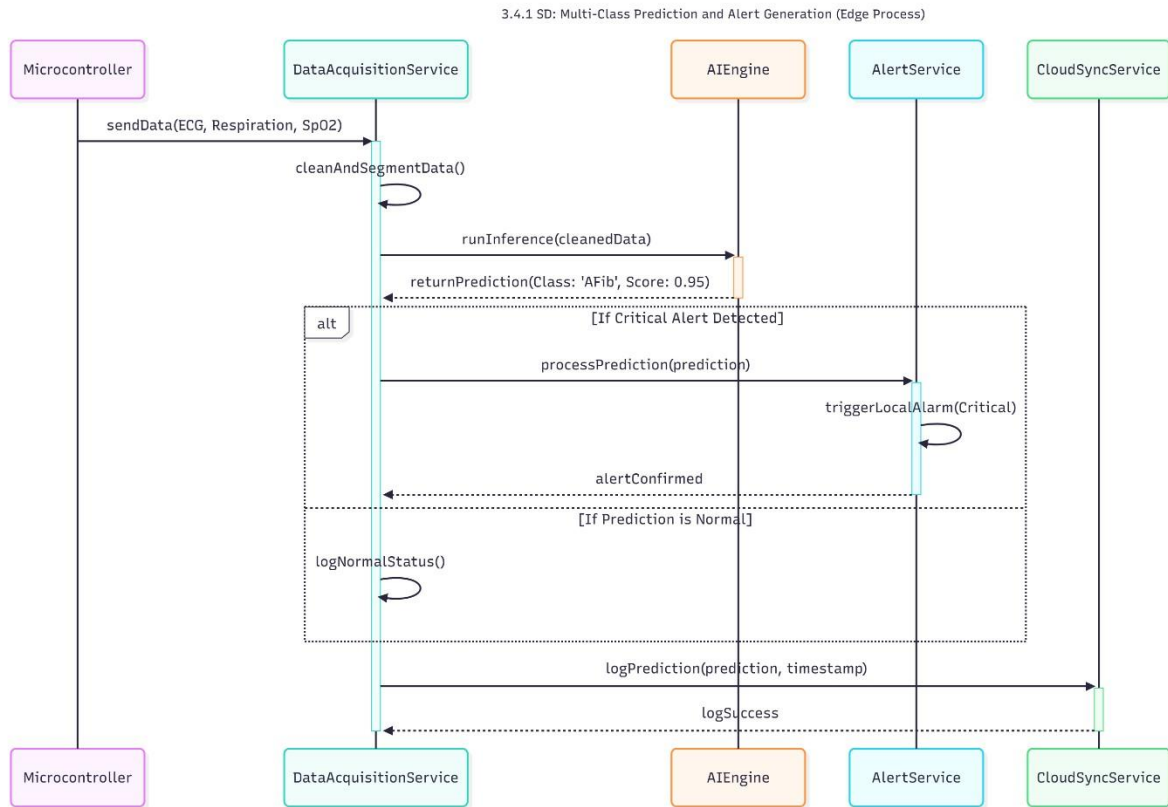
3.3. Class Diagram

Figure 25 3.3.1 Class Diagram



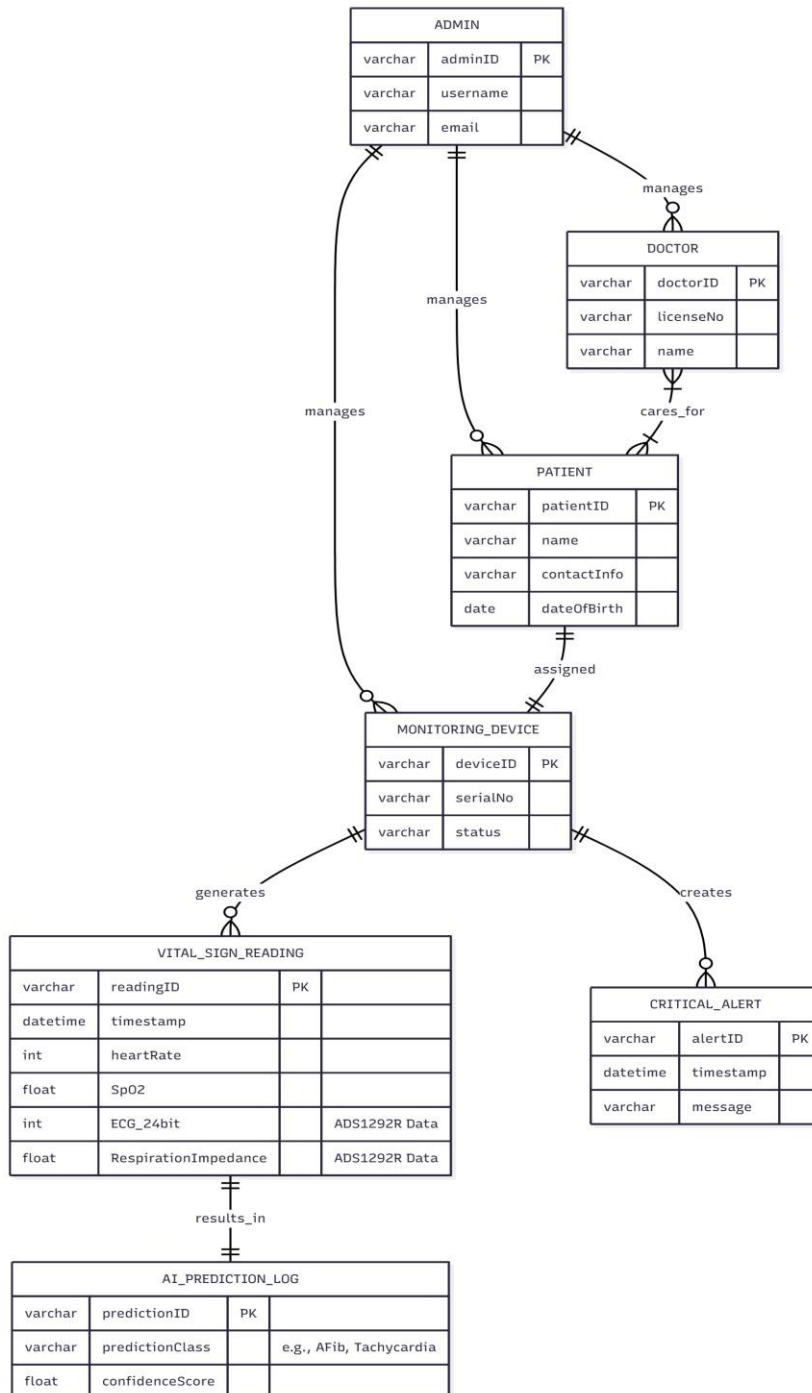
3.4. Sequence Diagram

Figure 26 3.4.1 Sequence Diagram



3.5. Entity Relationship Diagram

Figure 27 3.5.1 ERD

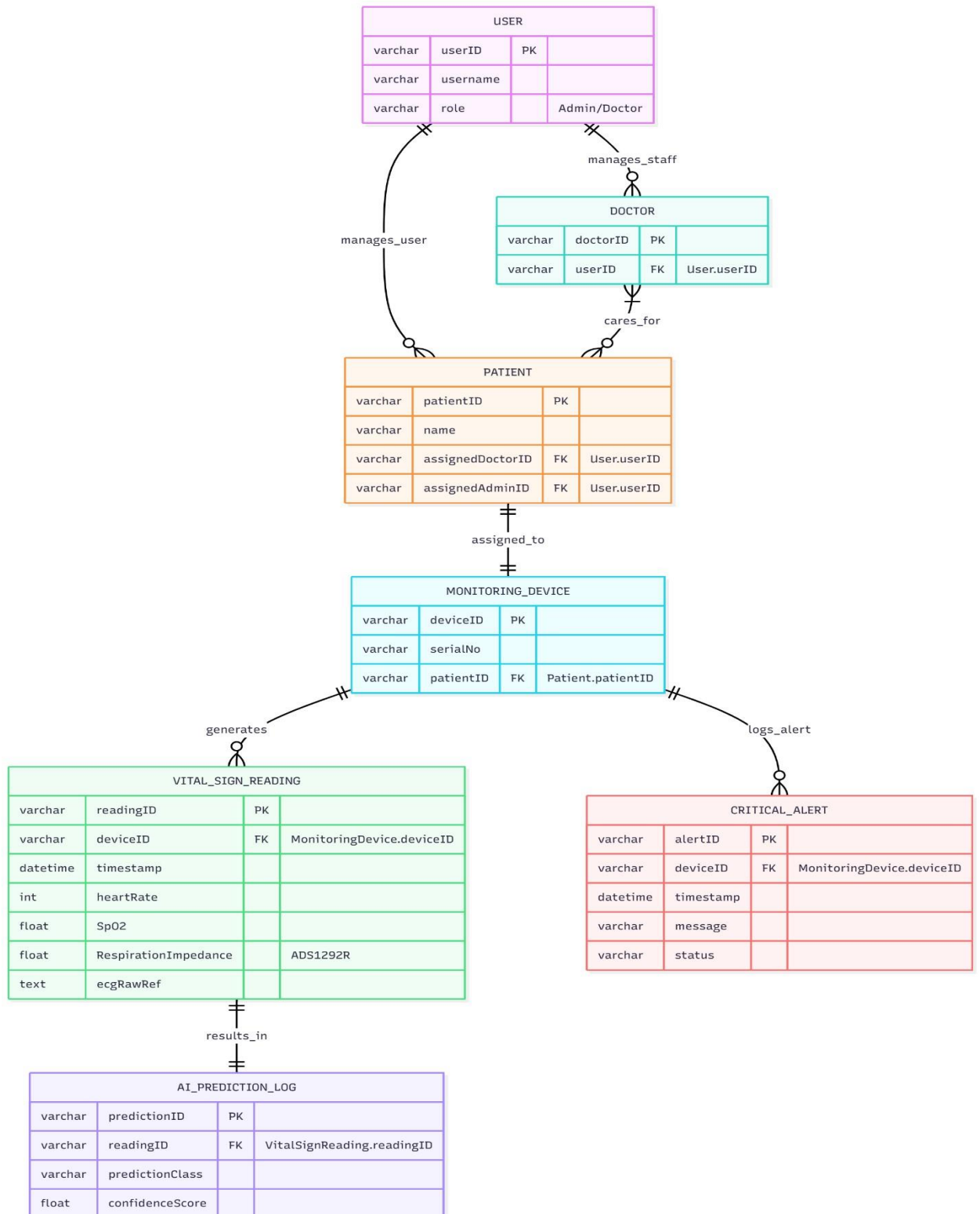


3.6. Database Schema

The database schema uses MySQL to store users, patients, doctors, devices, and vital data. Support for AI prediction logs is planned.

- **Tables:** Represented as boxes, each table corresponds to an entity in the system and includes a set of attributes. For this project, key tables include `PATIENT`, `VITAL_SIGN_READING`, and `AI_PREDICTION_LOG`.
- **Primary Key (PK):** A unique identifier for each record in a table. It ensures that no two rows have the same value for the primary key field (e.g., `patientID`).
- **Foreign Key (FK):** A field in one table that refers to the primary key in another table. It establishes a relationship between two tables, such as linking a `VITAL_SIGN_READING` record back to its `MONITORING_DEVICE`.
- **Relationships:** Lines connecting tables represent logical relationships such as One-to-One, One-to-Many, and Many-to-Many. These relationships define how data in one table relates to data in another.
- **Data Types and Constraints:** Columns include data types (e.g., `VARCHAR`, `INT`, `DATETIME`) and constraints (e.g., `NOT NULL` or explicit references to specialized types like **24-bit ECG** data) to ensure data validity and integrity across the system.

Figure 28 3.6.1 Database Schema



3.7. User Interface Design

The UI design outlines the key screens, prioritizing clarity and the immediate visibility of the new data streams.

Login page

Figure 29 3.7.1 login page

CUST Digihealth

in f Instagram SIGN UP

Log in

Email

Password

SIGN IN

CONTACT US ABOUT US FAQ

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Sign UP page :

Figure 30 3.7.2 signup page

CUST Digihealth

in f Instagram Sign in

Sign Up

Register a new membership

Enter User Name

Enter Email

- Account Type -

Password

SIGN UP

You already have a membership?

CONTACT US ABOUT US FAQ

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Figure 31 3.7.3 signup page

CUST Digihealth

Sign Up
Register a new membership

Enter User Name

Enter Email

- Account Type -
- Account Type -
Doctor
Patient

SIGN UP

You already have a membership?

CONTACT US ABOUT US FAQ

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Admin's Dashboard:

Figure 32 3.7.4 Admins Dashboard

gatewaylinks.org/digihealth/dashboard.php

CUST DIGIHEALTH

ADMIN (ADMIN)

Dashboard
Welcome to CUST Digihealth

Home / Dashboard

12 Doctors

8 Patients

18 Nursing Staff

Timeline

3 Wednesday
DECEMBER 2025

5pm PT-112
Abnormal Reading Received

4:35pm PT-112

New Patient List

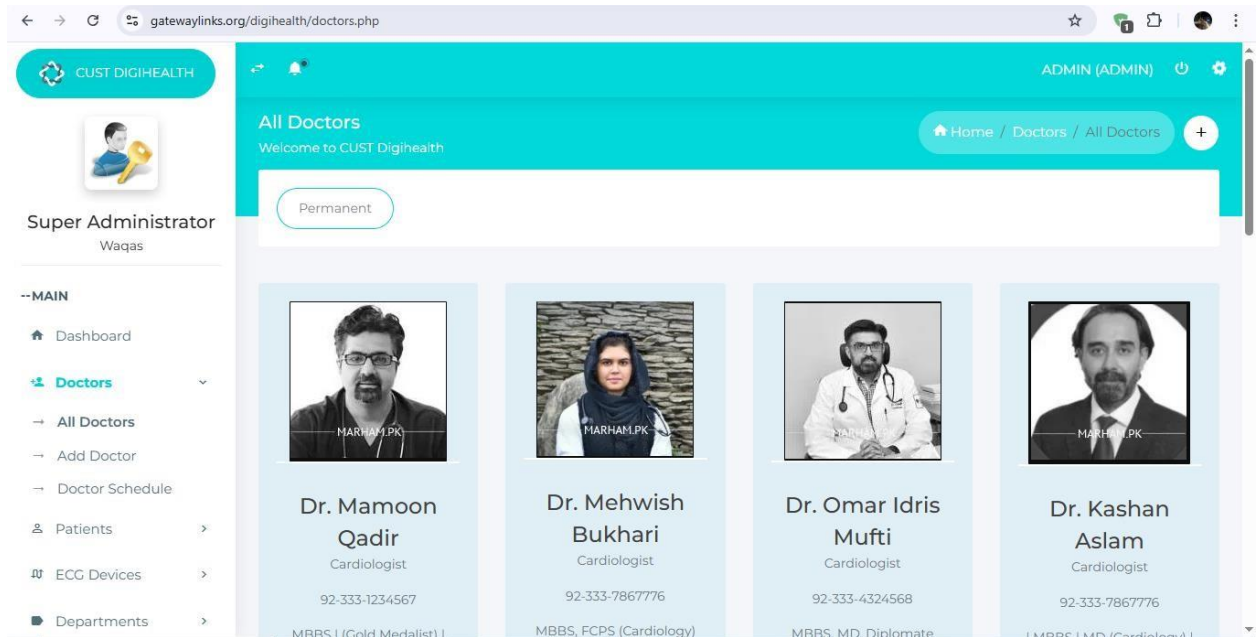
Patient ID	Associated Doctors	Att:
PT-100	Dr. Muhammad Omair Zahid (Cardiologist)	B0:I
PT-111	Dr. Fawad Nasrullah (Urologist), Prof. Dr. Umber Jallil (Gynecologist)	
PT-112	Dr. Kashan Aslam (Cardiologist), Dr. Usman Javaid (Gastroenterologist)	
PT-113	Dr. Mehwish Bukhari (Cardiologist)	B0:I
PT-23	Dr. Mamoon Qadir (Cardiologist), Dr. Mehwish Bukhari (Cardiologist)	d2:I

Super Administrator
Waqas

--MAIN

- Dashboard
- Doctors
- Patients
- ECG Devices
- Edge Devices Stats
- Departments

Figure 33 3.7.5 Admin Dashboard overview



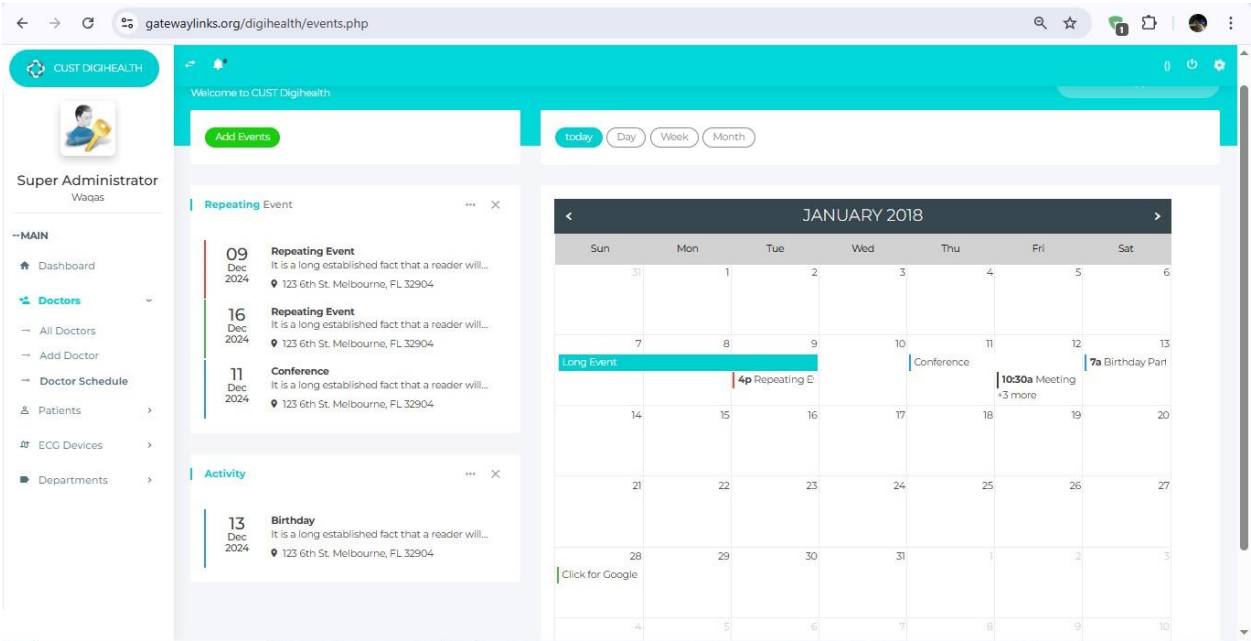
Add doctors :

Figure 34 3.7.6 Add doctors

The screenshot shows the 'Add Doctor' form in the CUST DigiHealth admin dashboard. The form is titled 'Create Doctor Profile' and includes sections for 'Doctor's Account Information' and 'Doctor Profile'. The 'Doctor's Account Information' section contains fields for User Name, Password, Email, and Confirm Password. The 'Doctor Profile' section contains fields for Full Name, Phone, Website Url, and Speciality, along with a file upload button for a profile picture. The form has 'Submit' and 'Cancel' buttons at the bottom.

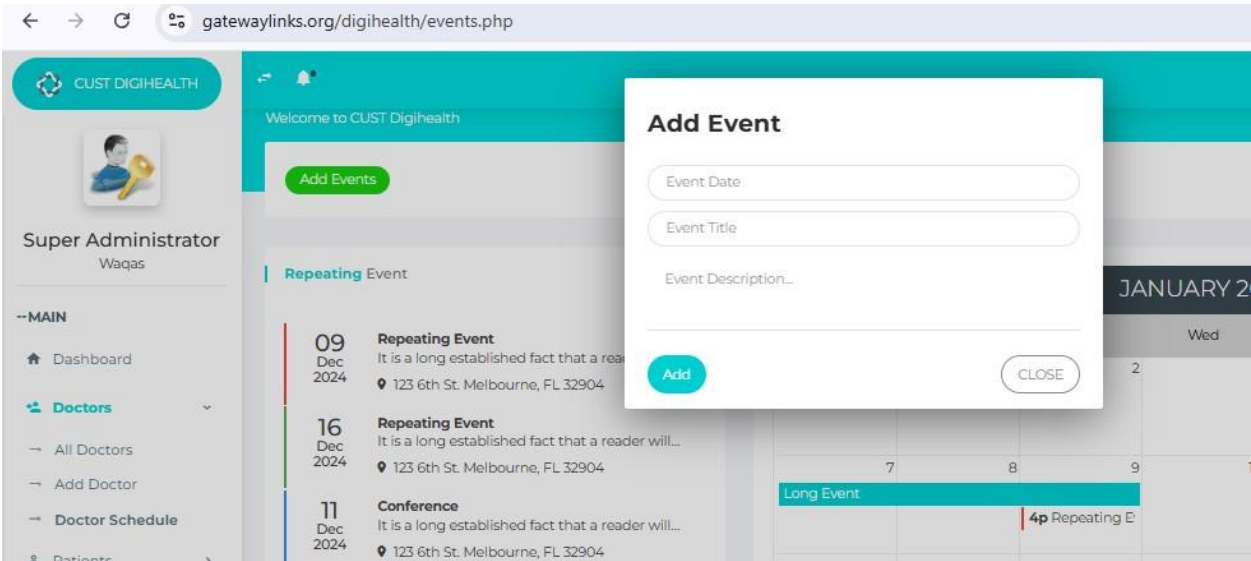
Doctor's Schedule ;

Figure 35 3.7.7 Doctors Schedule



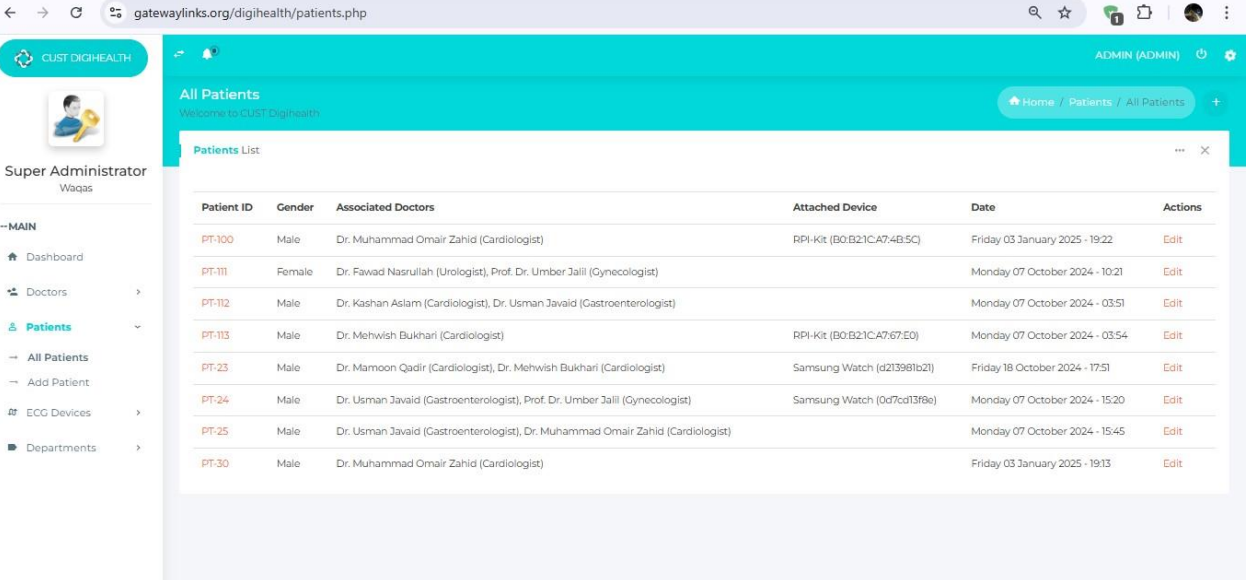
Add Events:

Figure 36 3.7.8 Add events



Pateint's :

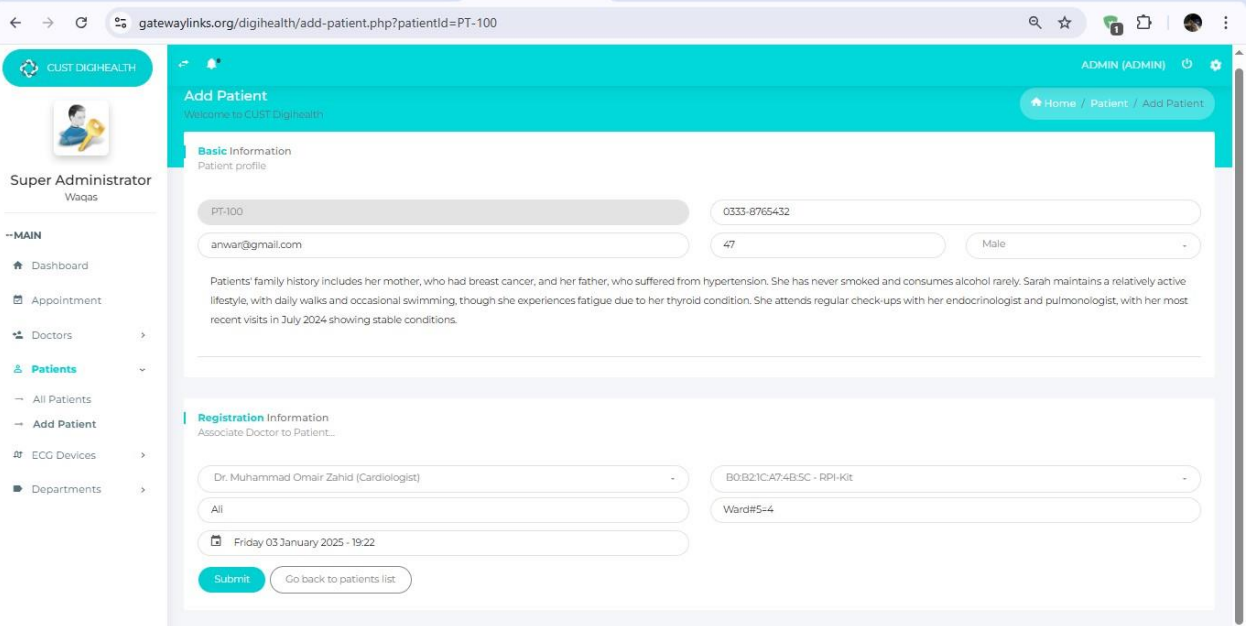
Figure 37 3.7.9 Patients view



Patient ID	Gender	Associated Doctors	Attached Device	Date	Actions
PT-100	Male	Dr. Muhammad Omair Zahid (Cardiologist)	RPI-Kit (B0B21CA7:4B:5C)	Friday 03 January 2025 - 19:22	Edit
PT-111	Female	Dr. Fawad Nasrullah (Urologist), Prof. Dr. Umber Jalil (Gynecologist)		Monday 07 October 2024 - 10:21	Edit
PT-112	Male	Dr. Kashan Aslam (Cardiologist), Dr. Usman Javaid (Gastroenterologist)		Monday 07 October 2024 - 03:51	Edit
PT-113	Male	Dr. Mehwish Bukhari (Cardiologist)	RPI-Kit (B0B21CA7:67:E0)	Monday 07 October 2024 - 03:54	Edit
PT-23	Male	Dr. Mamoon Qadir (Cardiologist), Dr. Mehwish Bukhari (Cardiologist)	Samsung Watch (d213981b21)	Friday 18 October 2024 - 17:51	Edit
PT-24	Male	Dr. Usman Javaid (Gastroenterologist), Prof. Dr. Umber Jalil (Gynecologist)	Samsung Watch (0d7cd13f8e)	Monday 07 October 2024 - 15:20	Edit
PT-25	Male	Dr. Usman Javaid (Gastroenterologist), Dr. Muhammad Omair Zahid (Cardiologist)		Monday 07 October 2024 - 15:45	Edit
PT-30	Male	Dr. Muhammad Omair Zahid (Cardiologist)		Friday 03 January 2025 - 19:13	Edit

Edit patient :

Figure 38 3.7.10 Edit Patient



Add Patient
Welcome to CUST DigHealth

Basic Information
Patient profile

Patient ID: PT-100
Email: anwar@gmail.com
Phone: 0333-8765432
Age: 47
Gender: Male

Patients' family history includes her mother, who had breast cancer, and her father, who suffered from hypertension. She has never smoked and consumes alcohol rarely. Sarah maintains a relatively active lifestyle, with daily walks and occasional swimming, though she experiences fatigue due to her thyroid condition. She attends regular check-ups with her endocrinologist and pulmonologist, with her most recent visits in July 2024 showing stable conditions.

Registration Information
Associate Doctor to Patient...

Dr. Muhammad Omair Zahid (Cardiologist)
Device: B0B21CA7:4B:5C - RPI-Kit
Ward: Ward#5=4
Date: Friday 03 January 2025 - 19:22

[Submit](#) [Go back to patients list](#)

Pateint's Profile :

Figure 39 3.7.11 Patients Profile

gatewaylinks.org/digihealth/Patient-Profile.php?patientId=PT-113

CUST DIGIHEALTH ADMIN (ADMIN)

Home / Patients / Patient Profile

Patient Profile

Welcome to CUST Digihhealth

Patient ID: PT-113

Associated Doctors
Dr. Mehwish Bukhari (Cardiologist)

Nursing Staff
All

Ward Number
D-4

Admission / Checkup Date
Monday 07 October 2024 - 03:54

Email ID
john@gmail.com

Phone
92-333-1234567

Medical History
Patient has a history of hypothyroidism (diagnosed in 2012) and asthma (diagnosed in childhood). She is currently on Levothyroxine 75 mcg daily for thyroid regulation and uses an albuterol inhaler as needed for asthma management. Sarah had a hysterectomy in 2015 due to uterine fibroids, with a smooth recovery. She is allergic to ibuprofen, which causes severe stomach upset, and avoids nonsteroidal anti-inflammatory drugs. Sarah was hospitalized briefly in March 2023 for an asthma flare-up triggered by a respiratory infection. She was treated with steroids and discharged within two days.

Super Administrator
Waqas

--MAIN

- Dashboard
- Appointment
- Doctors
- Patients**
 - All Patients
 - Add Patient
 - Patient Profile**
- ECG Devices
- Departments

Add Patient's :

Figure 40 3.7.12 add patients

gatewaylinks.org/digihealth/add-patient.php

CUST DIGIHEALTH ADMIN (ADMIN)

Home / Patient / Add Patient

Add Patient

Welcome to CUST Digihhealth

Basic Information
Patient profile

Patient Id:

Phone No.:

Enter Email:

Age:

Gender:

Description / Medical history:

Registration Information
Associate Doctor to Patient...

Nothing selected

Select Device:

Staff on Duty:

Ward No.:

Please choose date & time...

Submit **Go back to patients list**

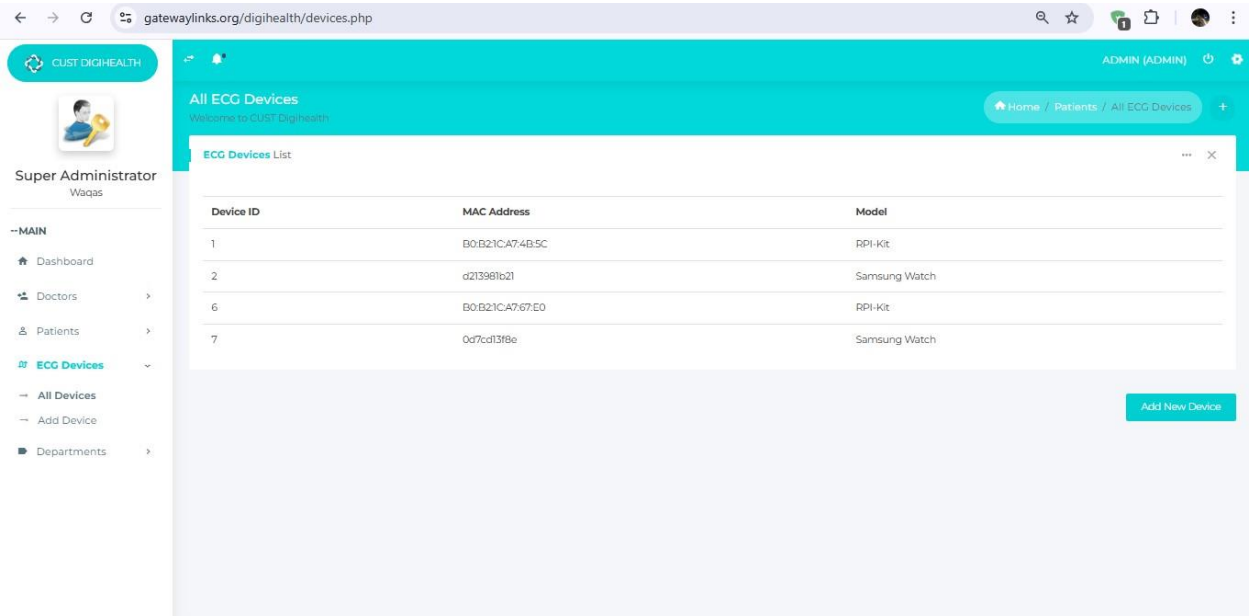
Super Administrator
Waqas

--MAIN

- Dashboard
- Appointment
- Doctors
- Patients**
 - All Patients
 - Add Patient**
 - ECG Devices
 - Departments

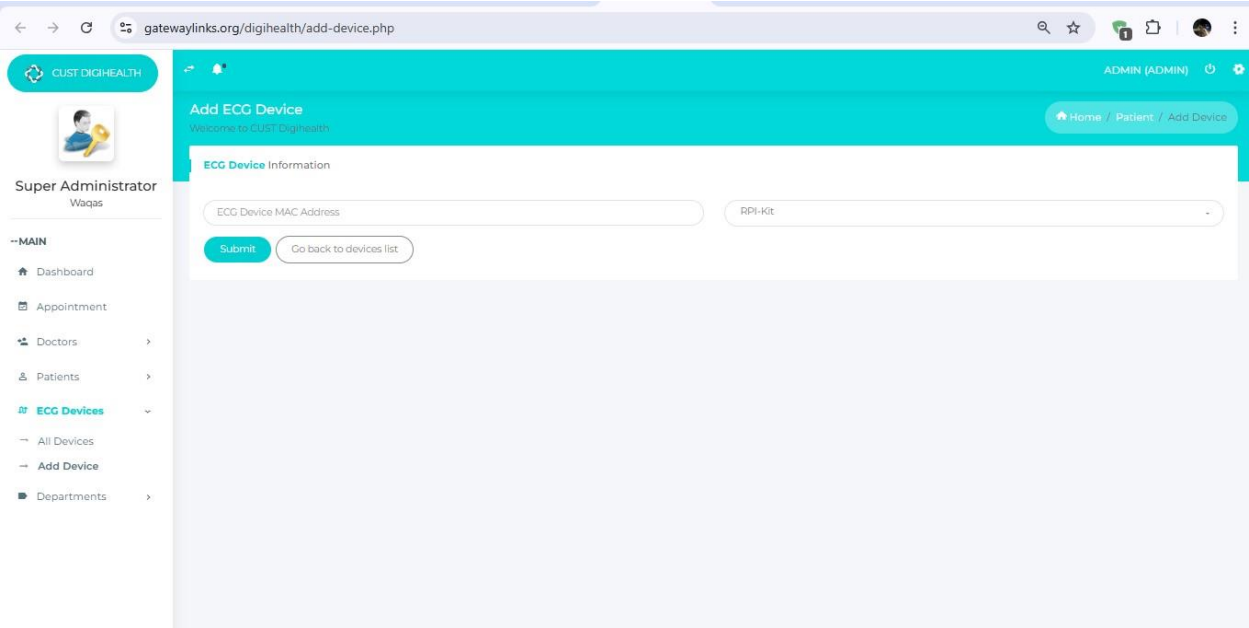
ECG Devices:

Figure 41 3.7.13 ECG Devices



Add Devices:

Figure 42 3.7.14 Add devices



Doctor's Dashboard:

Figure 43 3.7.15 Doctors Dashboard

The screenshot shows a web browser at the URL `gatewaylinks.org/digihealth/patients.php?status=Doctor%20login%20successful`. The page has a teal header with the user name "OMAIR (DOCTOR)" and a settings icon. Below the header, there's a navigation bar with "Home / Patients / All Patients". The main content area is titled "All Patients" and "Welcome to CUST Digihealth". It features a "Patients List" table with the following data:

Patient ID	Gender	Associated Doctors	Attached Device	Date	Actions
PT-100	Male	Dr. Muhammad Omair Zahid (Cardiologist)	RPI-Kit (B0:B2:1C:A7:4B:5C)	Friday 03 January 2025 - 19:22	Edit
PT-25	Male	Dr. Usman Javaid (Gastroenterologist), Dr. Muhammad Omair Zahid (Cardiologist)		Monday 07 October 2024 - 15:45	Edit
PT-30	Male	Dr. Muhammad Omair Zahid (Cardiologist)		Friday 03 January 2025 - 19:13	Edit

A green notification bar at the bottom right says "Doctor login successful".

Figure 44 3.7.16 Doctors edit patients

The screenshot shows the "Add Patient" form in the Doctor's Dashboard. The URL is `gatewaylinks.org/digihealth/add-patient.php?patientId=PT-100`. The page has a teal header with "OMAIR (DOCTOR)" and a settings icon. The navigation bar shows "Home / Patient / Add Patient". The form is titled "Add Patient" and "Welcome to CUST Digihealth". It is divided into two sections: "Basic Information" and "Registration Information".

Basic Information
Patient profile:

PT-100
anwar@gmail.com
0333-8765432
47
Male

Patients' family history includes her mother, who had breast cancer, and her father, who suffered from hypertension. She has never smoked and consumes alcohol rarely. Sarah maintains a relatively active lifestyle, with daily walks and occasional swimming, though she experiences fatigue due to her thyroid condition. She attends regular check-ups with her endocrinologist and pulmonologist, with her most recent visits in July 2024 showing stable conditions.

Registration Information
Associate Doctor to Patient...

Dr. Muhammad Omair Zahid (Cardiologist)
All
Friday 03 January 2025 - 19:22
B0:B2:1C:A7:4B:5C - RPI-Kit
Ward#5-4

Buttons: Submit, Go back to patients list

Figure 45 3.7.17 Doctor can add patients

gatewaylinks.org/digihealth/add-patient.php

OMAIR (DOCTOR)

[Home](#) / [Patient](#) / [Add Patient](#)

Add Patient

Welcome to CUST Digihealth

Basic Information

Patient profile

Patient Id:

Phone No.:

Enter Email:

Age:

Gender:

Description / Medical history:

Registration Information

Associate Doctor to Patient...

Nothing selected

Select Device

Staff on Duty:

Ward No.:

Please choose date & time...

[Submit](#) [Go back to patients list](#)

Patient's Dashboard

Figure 46 3.7. 18 Patients Dashboard

gatewaylinks.org/digihealth/Patient-Profile.php?patientId=PT-100

PT-100 (PATIENT)

[Home](#) / [Patients](#) / [Patient Profile](#)

Patient Profile

Welcome to CUST Digihealth

Patient ID: PT-100

Associated Doctors
Dr. Muhammad Omair Zahid (Cardiologist)

Nursing Staff
All

Ward Number
Ward#5-4

Admission / Checkup Date
Friday 03 January 2025 - 19:22

Email ID
arwar@gmail.com

Phone
0333-8765432

Medical History

Patients' family history includes her mother, who had breast cancer, and her father, who suffered from hypertension. She has never smoked and consumes alcohol rarely. Sarah maintains a relatively active lifestyle, with daily walks and occasional swimming, though she experiences fatigue due to her thyroid condition. She attends regular check-ups with her endocrinologist and pulmonologist, with her most recent visits in July 2024 showing stable conditions.

- Normal ECG Received on 2025-11-14 02:51:38 (from device: B0:B21CA7:4B:5C)
- Normal ECG Received on 2025-11-14 02:51:36 (from device: B0:B21CA7:4B:5C)
- Normal ECG Received on 2025-11-14 02:38:06 (from device: B0:B21CA7:4B:5C)
- Normal ECG Received on 2025-11-11 01:52:23 (from device: B0:B21CA7:4B:5C)
- Normal ECG Received on 2025-11-05 14:59:29 (from device: B0:B21CA7:4B:5C)
- Normal ECG Received on 2025-11-05 14:39:52 (from device: B0:B21CA7:4B:5C)
- Normal ECG Received on 2025-01-03 19:57:57 (from device: B0:B21CA7:4B:5C)
- Abnormal ECG Received on 2025-01-03 19:50:57 (from device: B0:B21CA7:4B:5C)

Chapter 4: Implementation

The system is implemented using PHP and MySQL. The backend manages authentication, role-based access control, patient records, device registration, and dashboard rendering. The api/ module handles communication with monitoring devices. No AI inference or ADS1292R preprocessing is included in the current phase.

4.1. Development Environment

The development of the AI-Powered Cardiovascular Patient Monitoring System was carried out using PHP as the backend programming language and MySQL as the database management system. XAMPP was used as the local development environment, providing Apache server and MySQL database services. The frontend interfaces were developed using HTML, CSS, and JavaScript. Raspberry Pi devices are supported as monitoring gateways for transmitting physiological data to the web application.

4.2. System Folder Structure

The system follows a modular folder structure. The api directory handles communication between monitoring devices and the backend system. The assets directory contains CSS, JavaScript, and library files used for the user interface. Backend logic for authentication, patient management, and data handling is implemented using PHP scripts.

4.3. Backend Implementation (PHP)

The backend of the system is implemented using PHP. It handles user authentication, role-based access control, patient profile management, device registration, and retrieval of monitoring data. Sessions are used to maintain secure user access across the system.

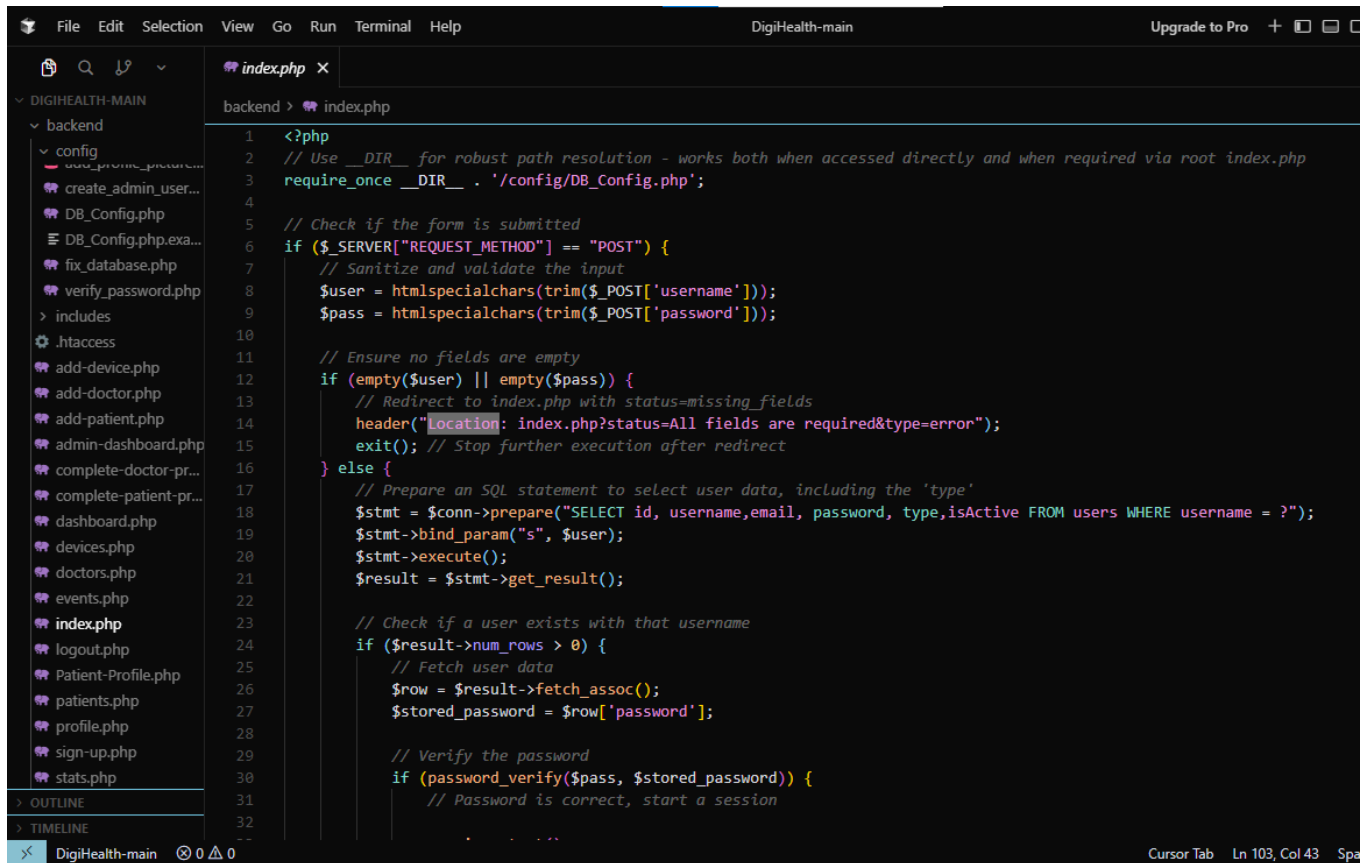
4.4. Database Implementation (MySQL)

MySQL is used as the database management system. The database stores user credentials, patient profiles, doctor records, device information, and physiological readings. The database schema is designed to support future expansion for AI prediction logs.

4.5. User Interface Implementation

4.5.1 Admin Dashboard Section

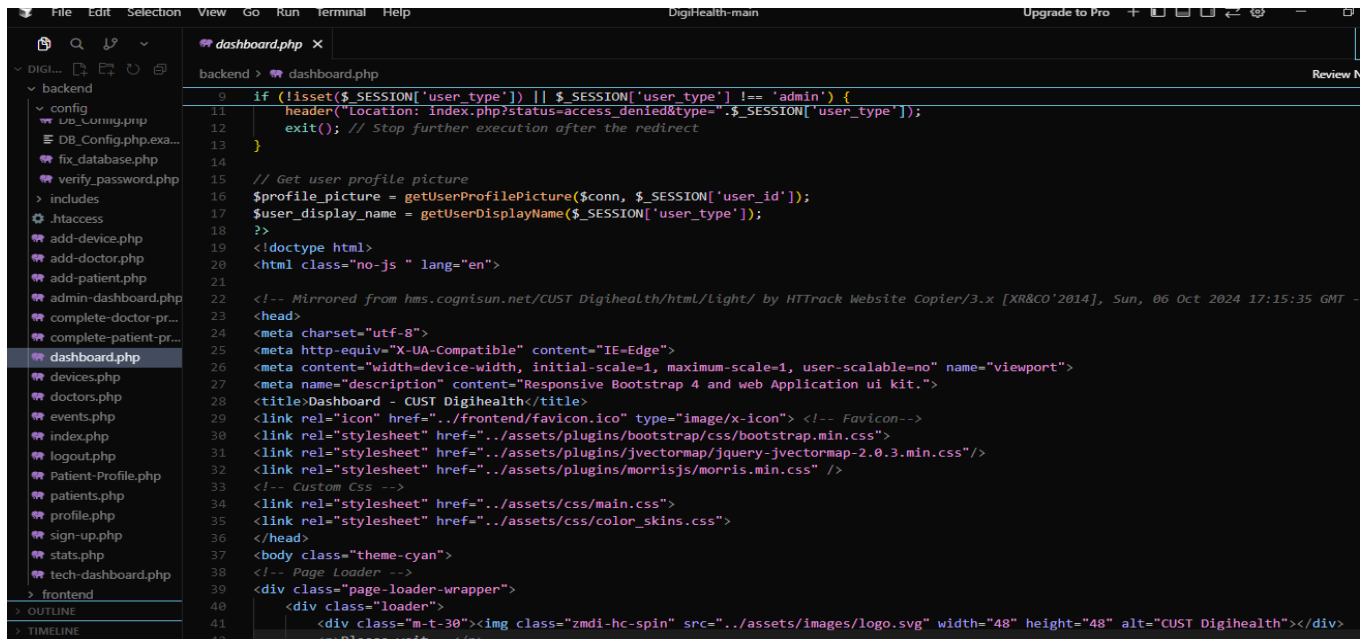
Admin Login Page



```
1 <?php
2 // Use __DIR__ for robust path resolution - works both when accessed directly and when required via root index.php
3 require_once __DIR__ . '/config/DB_Config.php';
4
5 // Check if the form is submitted
6 if ($_SERVER["REQUEST_METHOD"] == "POST") {
7     // Sanitize and validate the input
8     $user = htmlspecialchars(trim($_POST['username']));
9     $pass = htmlspecialchars(trim($_POST['password']));
10
11     // Ensure no fields are empty
12     if (empty($user) || empty($pass)) {
13         // Redirect to index.php with status=missing_fields
14         header("Location: index.php?status=All fields are required&type=error");
15         exit(); // Stop further execution after redirect
16     } else {
17         // Prepare an SQL statement to select user data, including the 'type'
18         $stmt = $conn->prepare("SELECT id, username, email, password, type, isActive FROM users WHERE username = ?");
19         $stmt->bind_param("s", $user);
20         $stmt->execute();
21         $result = $stmt->get_result();
22
23         // Check if a user exists with that username
24         if ($result->num_rows > 0) {
25             // Fetch user data
26             $row = $result->fetch_assoc();
27             $stored_password = $row['password'];
28
29             // Verify the password
30             if (password_verify($pass, $stored_password)) {
31                 // Password is correct, start a session
32                 // ...
33             }
34         }
35     }
36 }
```

Figure 47 4.5.1 Admin login page

Admin Dashboard

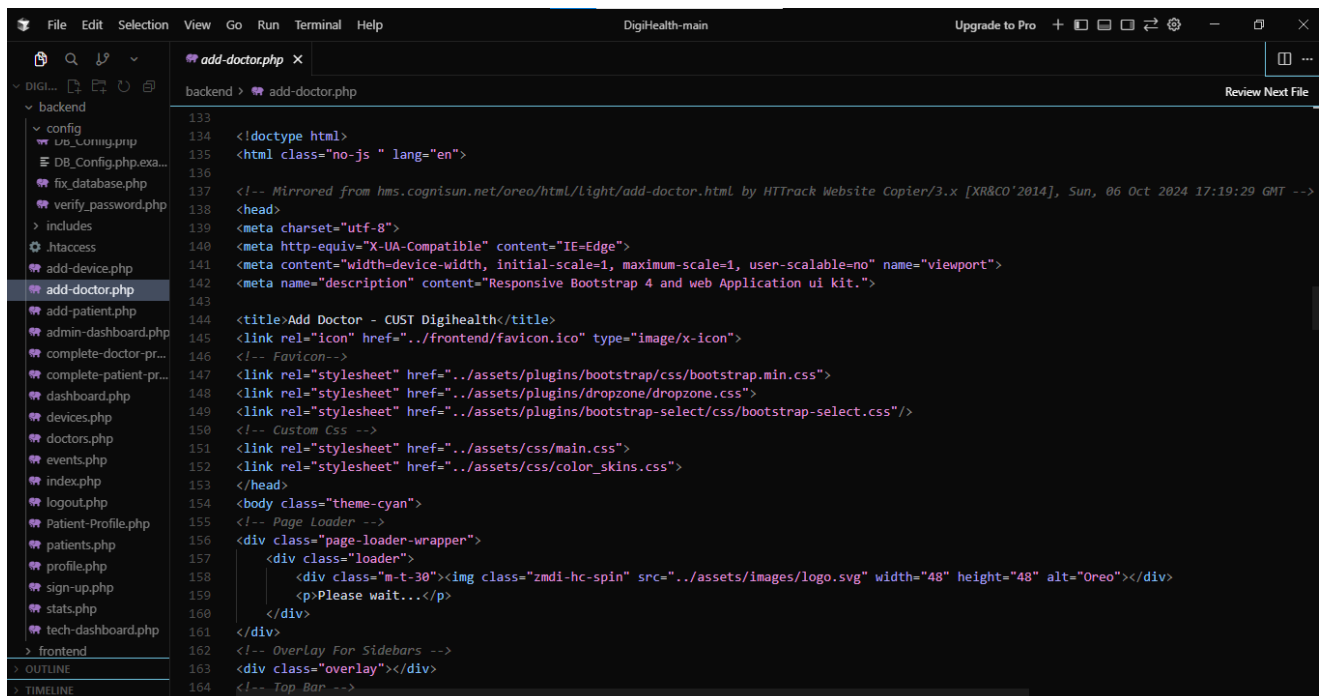


The screenshot shows a code editor with a file explorer on the left and a code editor on the right. The file explorer shows a project structure with a 'backend' directory containing various PHP files. The 'dashboard.php' file is selected. The code editor shows the following PHP code:

```
9 if (!isset($_SESSION['user_type']) || $_SESSION['user_type'] != 'admin') {
11     header("Location: index.php?status=access_denied&type=".$_SESSION['user_type']);
12     exit(); // Stop further execution after the redirect
13 }
14
15 // Get user profile picture
16 $profile_picture = getUserProfilePicture($conn, $_SESSION['user_id']);
17 $user_display_name = getUserDisplayName($_SESSION['user_type']);
18
19 <!doctype html>
20 <html class="no-js" lang="en">
21
22 <!-- Mirrored from hms.cognisun.net/CUST Digihealth/html/Light/ by HTTrack Website Copier/3.x [XR&CO'2014], Sun, 06 Oct 2024 17:15:35 GMT -->
23 <head>
24 <meta charset="utf-8">
25 <meta http-equiv="X-UA-Compatible" content="IE=edge">
26 <meta content="width=device-width, initial-scale=1, maximum-scale=1, user-scalable=no" name="viewport">
27 <meta name="description" content="Responsive Bootstrap 4 and web Application ui kit.">
28 <title>Dashboard - CUST Digihealth</title>
29 <link rel="icon" href="../frontend/favicon.ico" type="image/x-icon"> <!-- Favicon -->
30 <link rel="stylesheet" href="../assets/plugins/bootstrap/css/bootstrap.min.css">
31 <link rel="stylesheet" href="../assets/plugins/jvectormap/jquery-jvectormap-2.0.3.min.css"/>
32 <link rel="stylesheet" href="../assets/plugins/morrisjs/morris.min.css" />
33 <!-- Custom CSS -->
34 <link rel="stylesheet" href="../assets/css/main.css">
35 <link rel="stylesheet" href="../assets/css/color_skins.css">
36 </head>
37 <body class="theme-cyan">
38 <!-- Page Loader -->
39 <div class="page-loader-wrapper">
40 <div class="loader">
41 <div class="m-t-30"></div>
```

Figure 48 4.5.2 admin dashboard

Add Doctor Page

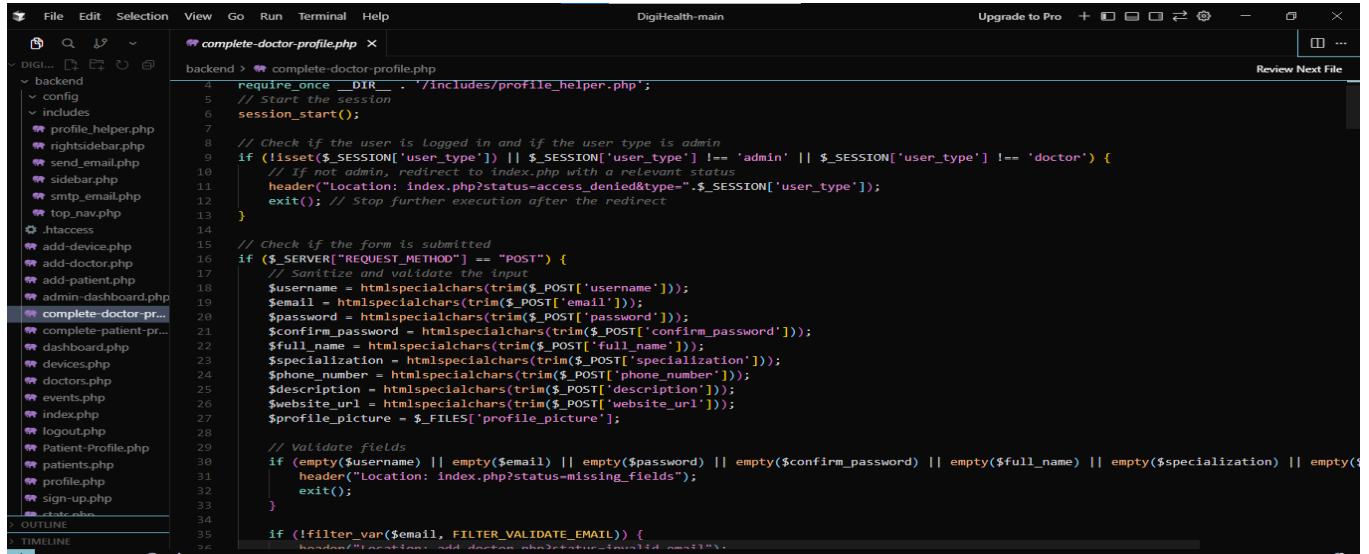


The screenshot shows a code editor with a file explorer on the left and a code editor on the right. The file explorer shows a project structure with a 'backend' directory containing various PHP files. The 'add-doctor.php' file is selected. The code editor shows the following PHP code:

```
133 <!doctype html>
134 <html class="no-js" lang="en">
135
136 <!-- Mirrored from hms.cognisun.net/oreo/html/Light/add-doctor.html by HTTrack Website Copier/3.x [XR&CO'2014], Sun, 06 Oct 2024 17:19:29 GMT -->
137 <head>
138 <meta charset="utf-8">
139 <meta http-equiv="X-UA-Compatible" content="IE=edge">
140 <meta content="width=device-width, initial-scale=1, maximum-scale=1, user-scalable=no" name="viewport">
141 <meta name="description" content="Responsive Bootstrap 4 and web Application ui kit.">
142
143 <title>Add Doctor - CUST Digihealth</title>
144 <link rel="icon" href="../frontend/favicon.ico" type="image/x-icon">
145 <!-- Favicon -->
146 <link rel="stylesheet" href="../assets/plugins/bootstrap/css/bootstrap.min.css">
147 <link rel="stylesheet" href="../assets/plugins/dropzone/dropzone.css">
148 <link rel="stylesheet" href="../assets/plugins/bootstrap-select/css/bootstrap-select.css"/>
149 <!-- Custom CSS -->
150 <link rel="stylesheet" href="../assets/css/main.css">
151 <link rel="stylesheet" href="../assets/css/color_skins.css">
152 </head>
153 <body class="theme-cyan">
154 <!-- Page Loader -->
155 <div class="page-loader-wrapper">
156 <div class="loader">
157 <div class="m-t-30"></div>
158 <p>Please wait...</p>
159 </div>
160 </div>
161 <!-- Overlay For Sidebars -->
162 <div class="overlay"></div>
163 <!-- Top Bar -->
```

Figure 49 4.5.3 Add doctor page

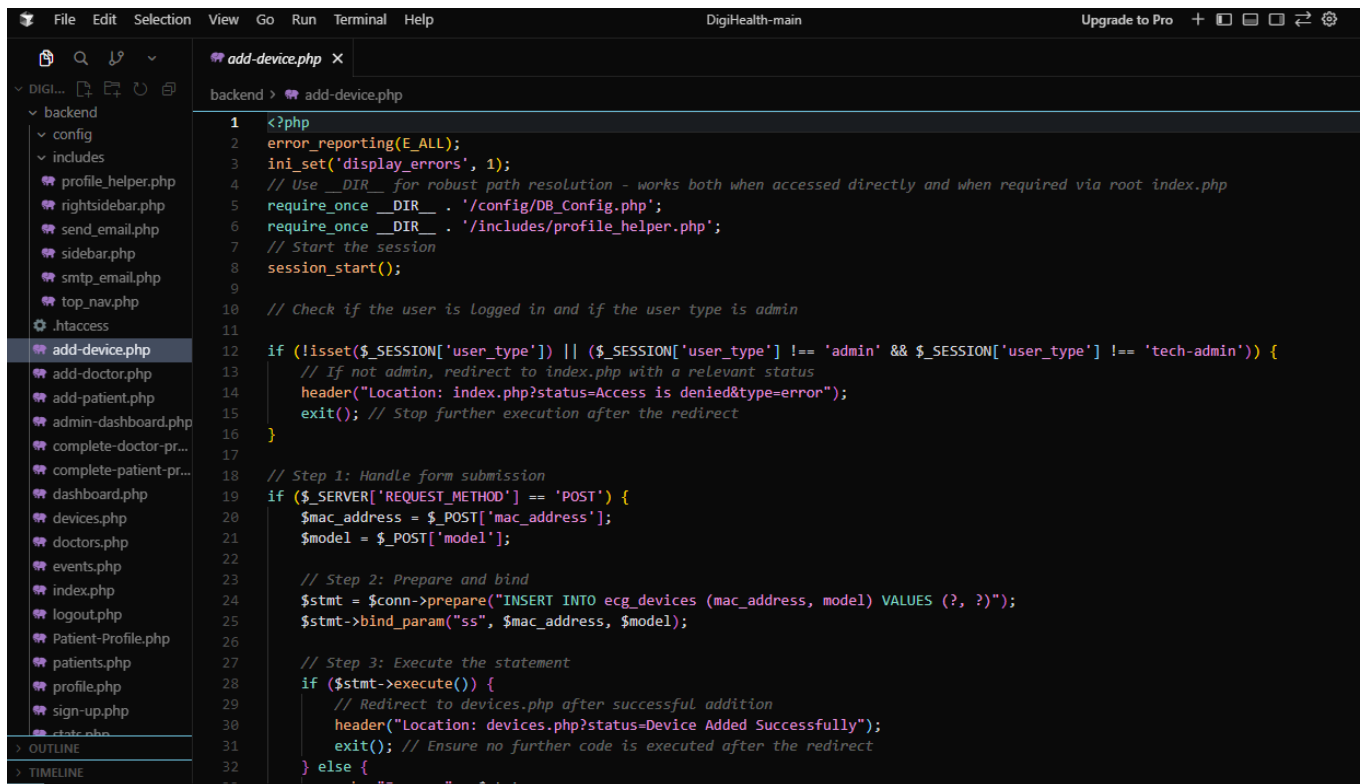
View All Doctors



```
4 require_once __DIR__ . '/includes/profile_helper.php';
5 // Start the session
6 session_start();
7
8 // Check if the user is logged in and if the user type is admin
9 if (!isset($_SESSION['user_type']) || $_SESSION['user_type'] !== 'admin' || $_SESSION['user_type'] !== 'doctor') {
10     // If not admin, redirect to index.php with a relevant status
11     header("Location: index.php?status=access_denied&type=" . $_SESSION['user_type']);
12     exit(); // Stop further execution after the redirect
13 }
14
15 // Check if the form is submitted
16 if ($_SERVER['REQUEST_METHOD'] === "POST") {
17     // Sanitize and validate the input
18     $username = htmlspecialchars(trim($_POST['username']));
19     $email = htmlspecialchars(trim($_POST['email']));
20     $password = htmlspecialchars(trim($_POST['password']));
21     $confirm_password = htmlspecialchars(trim($_POST['confirm_password']));
22     $full_name = htmlspecialchars(trim($_POST['full_name']));
23     $specialization = htmlspecialchars(trim($_POST['specialization']));
24     $phone_number = htmlspecialchars(trim($_POST['phone_number']));
25     $description = htmlspecialchars(trim($_POST['description']));
26     $website_url = htmlspecialchars(trim($_POST['website_url']));
27     $profile_picture = $_FILES['profile_picture'];
28
29     // Validate fields
30     if (empty($username) || empty($email) || empty($password) || empty($confirm_password) || empty($full_name) || empty($specialization) || empty($phone_number) || empty($description) || empty($website_url) || empty($profile_picture)) {
31         header("Location: index.php?status=missing_fields");
32         exit();
33     }
34
35     if (!filter_var($email, FILTER_VALIDATE_EMAIL)) {
36         header("Location: index.php?status=invalid_email");
37         exit();
38     }
39 }
```

Figure 50 4.5.4 view doctors

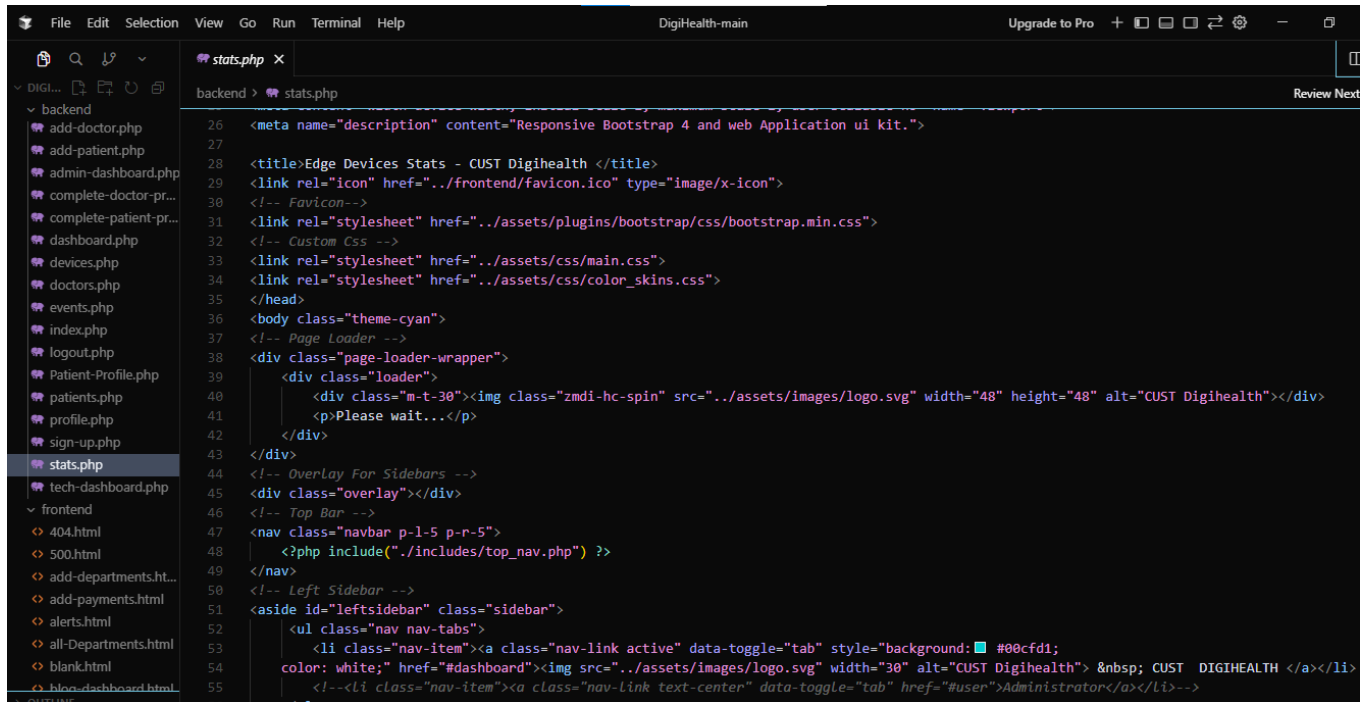
Add Device Page



```
1 <?php
2 error_reporting(E_ALL);
3 ini_set('display_errors', 1);
4 // Use __DIR__ for robust path resolution - works both when accessed directly and when required via root index.php
5 require_once __DIR__ . '/config/DB_Config.php';
6 require_once __DIR__ . '/includes/profile_helper.php';
7 // Start the session
8 session_start();
9
10 // Check if the user is logged in and if the user type is admin
11 if (!isset($_SESSION['user_type']) || ($_SESSION['user_type'] !== 'admin' && $_SESSION['user_type'] !== 'tech-admin')) {
12     // If not admin, redirect to index.php with a relevant status
13     header("Location: index.php?status=Access is denied&type=error");
14     exit(); // Stop further execution after the redirect
15 }
16
17 // Step 1: Handle form submission
18 if ($_SERVER['REQUEST_METHOD'] === 'POST') {
19     $mac_address = $_POST['mac_address'];
20     $model = $_POST['model'];
21
22     // Step 2: Prepare and bind
23     $stmt = $conn->prepare("INSERT INTO ecg_devices (mac_address, model) VALUES (?, ?)");
24     $stmt->bind_param("ss", $mac_address, $model);
25
26     // Step 3: Execute the statement
27     if ($stmt->execute()) {
28         // Redirect to devices.php after successful addition
29         header("Location: devices.php?status=Device Added Successfully");
30         exit(); // Ensure no further code is executed after the redirect
31     } else {
32         // Handle error
33     }
34 }
```

Figure 51 4.5.5 add device page

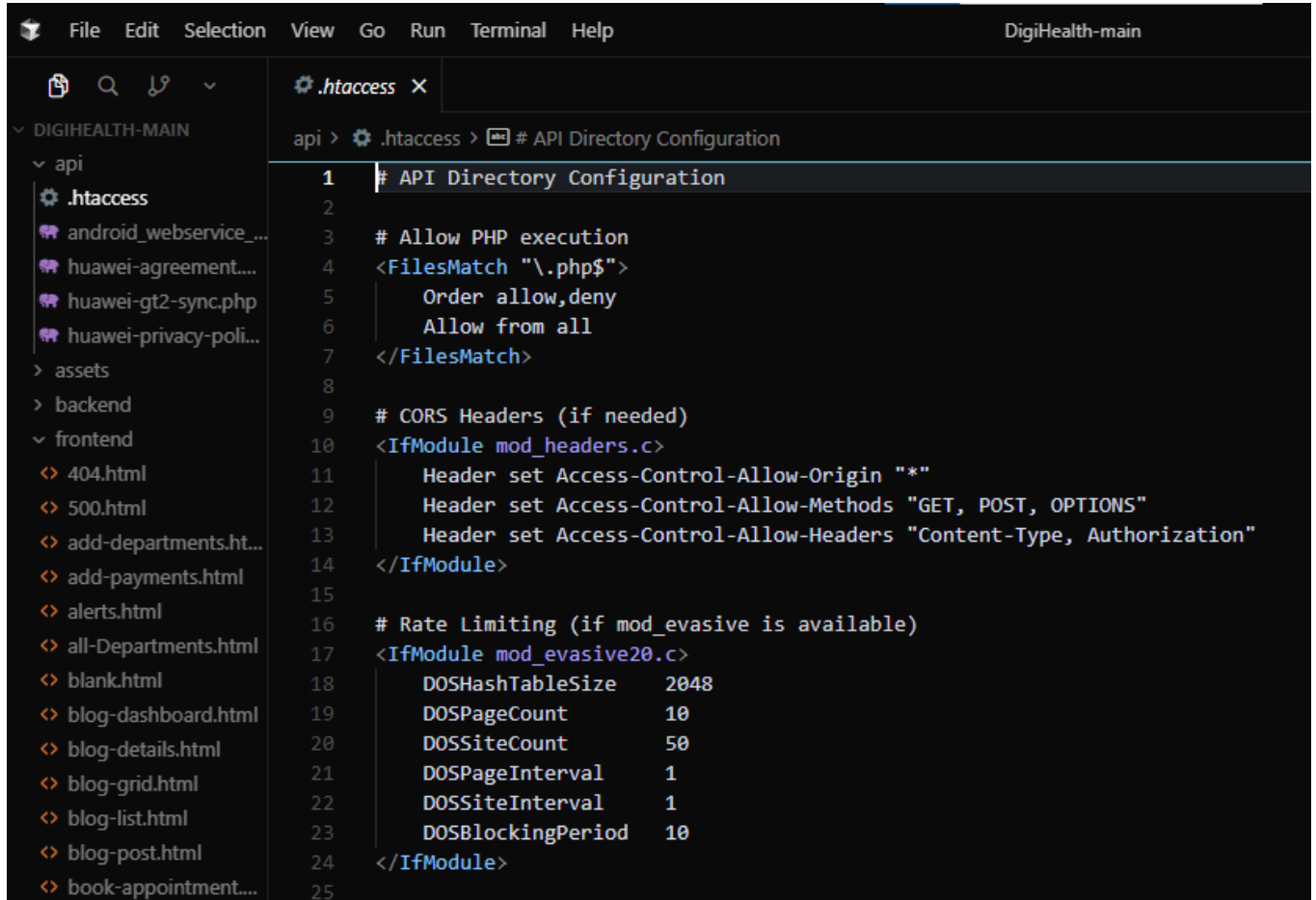
Medical History Page



```
File Edit Selection View Go Run Terminal Help DigiHealth-main Upgrade to Pro + [Icons] - [Icons]
stats.php X
backend > stats.php Review Next
26 <meta name="description" content="Responsive Bootstrap 4 and web Application ui kit.">
27
28 <title>Edge Devices Stats - CUST Digihealth </title>
29 <link rel="icon" href="../frontend/favicon.ico" type="image/x-icon">
30 <!-- Favicon -->
31 <link rel="stylesheet" href="../assets/plugins/bootstrap/css/bootstrap.min.css">
32 <!-- Custom Css -->
33 <link rel="stylesheet" href="../assets/css/main.css">
34 <link rel="stylesheet" href="../assets/css/color_skins.css">
35 </head>
36 <body class="theme-cyan">
37 <!-- Page Loader -->
38 <div class="page-loader-wrapper">
39   <div class="loader">
40     <div class="m-t-30"></div>
41     <p>Please wait...</p>
42   </div>
43 </div>
44 <!-- Overlay For Sidebars -->
45 <div class="overlay"></div>
46 <!-- Top Bar -->
47 <nav class="navbar p-1-5 p-r-5">
48   <?php include("../includes/top_nav.php") ?>
49 </nav>
50 <!-- Left Sidebar -->
51 <aside id="leftsidebar" class="sidebar">
52   <ul class="nav nav-tabs">
53     <li class="nav-item"><a class="nav-link active" data-toggle="tab" style="background-color: #00cfd1; color: white;" href="#dashboard"> &nbsp; CUST DIGIHEALTH </a></li>
54     <!--<li class="nav-item"><a class="nav-link text-center" data-toggle="tab" href="#user">Administrator</a></li>-->
55   </ul>
56 </aside>
```

Figure 52 4.5.6 medical history

API Module Structure



```
1 # API Directory Configuration
2
3 # Allow PHP execution
4 <FilesMatch "\.php$">
5     Order allow,deny
6     Allow from all
7 </FilesMatch>
8
9 # CORS Headers (if needed)
10 <IfModule mod_headers.c>
11     Header set Access-Control-Allow-Origin "*"
12     Header set Access-Control-Allow-Methods "GET, POST, OPTIONS"
13     Header set Access-Control-Allow-Headers "Content-Type, Authorization"
14 </IfModule>
15
16 # Rate Limiting (if mod_evasive is available)
17 <IfModule mod_evasive20.c>
18     DOSHashTableSize    2048
19     DOSPageCount        10
20     DOSSiteCount        50
21     DOSPageInterval     1
22     DOSSiteInterval     1
23     DOSBlockingPeriod   10
24 </IfModule>
25
```

Figure 53 4.5.7 API Modules

The API module handles communication between monitoring devices and the web application.

4.6. Device Integration

The system supports Raspberry Pi devices as monitoring gateways. Devices can be registered and managed through the admin interface. Sensor-level signal preprocessing and medical-grade ECG acquisition using ADS1292R are planned for future implementation.

Chapter 5: Software Testing

5.1. Testing Methodology

The system was tested using **Black-Box Testing methodology**. In black-box testing, the internal code structure is not considered. Instead, the system is evaluated based on input-output behavior and requirement compliance.

Test cases were designed according to the functional requirements defined in Chapter 2. Each test case includes:

- Test Case ID
- Feature Tested
- Preconditions
- Test Steps
- Expected Result
- Actual Result
- Status

This approach ensures that the system behaves correctly from a user's perspective.

5.2. Test Environment

Testing was performed on a local development environment using XAMPP and MySQL. The system was accessed using standard web browsers. Raspberry Pi devices were tested for connectivity and data transmission through APIs.

Testing was performed under the following environment:

- **Web Server:** PHP Development Server (localhost:8000)
- **Database:** MySQL (gatewayl_ecg_db)
- **Development Stack:** XAMPP (Apache + MySQL)
- **Backend Language:** PHP

- **Frontend Technologies:** HTML, CSS, JavaScript
- **Browser:** Google Chrome
- **Tool:** Vs Code / Cursor AI

The database schema was imported successfully, and the configuration file (DB_Config.php) was verified for correct credentials before testing.

5.3. Test Cases

Functional testing verifies that each implemented module performs according to system requirements.

5.3.1 Test Case 1 – User Login

Table 19 5.3.1 TC

Field	Description
Test Case ID	TC-01
Feature Tested	User Authentication
Preconditions	User account exists and is active
Test Steps	1. Open login page 2. Enter valid username and password 3. Click Login
Expected Result	User is redirected to the appropriate dashboard
Actual Result	Successful redirection based on role
Status	Pass

Additional validation:

- Invalid password → Error message displayed
- Inactive user → Access denied

5.3.2 Test Case 2 – Role-Based Access Control (RBAC)

Table 20 5.3.2 TC 2

Field	Description
Test Case ID	TC-02
Feature Tested	Role-Based Authorization
Preconditions	User logged in
Test Steps	1. Login as Admin 2. Attempt to access Doctor page 3. Login as Doctor 4. Attempt to access Admin panel
Expected Result	Unauthorized pages are restricted
Actual Result	Access correctly restricted
status	Pass

The system successfully prevents unauthorized access and redirects users to login if sessions are not valid.

5.3.3 Test Case 3 – Add Patient

Table 21 5.3.3 TC 3

Field	Description
Test Case ID	TC-03
Feature Tested	Patient Registration
Preconditions	Doctor logged in
Test Steps	1. Navigate to Add Patient page 2. Enter patient details 3. Submit form
Expected Result	Patient record stored in database
Actual Result	Patient successfully added
Status	Pass

Form validation was also tested for missing required fields.

5.3.4 Test Case 4 – Device Registration

Table 22 5.3. 4 TC 4

Field	Description
Test Case ID	TC-04
Feature Tested	Device Registration
Preconditions	Admin logged in
Test Steps	1. Navigate to Add Device page 2. Enter device details 3. Submit
Expected Result	Device stored in database
Actual Result	Device successfully registered
Status	Pass

Duplicate device IDs were tested and correctly rejected.

5.3.5 Test Case 5 – Dashboard Loading

Table 23 5.3.5 TC 5

Field	Description
Test Case ID	TC-05
Feature Tested	Dashboard Visualization
Preconditions	User logged in
Test Steps	1. Login to system 2. Redirect to dashboard
Expected Result	Dashboard loads and displays relevant data
Actual Result	Dashboard displayed correctly
Status	Pass

5.4. Testing Summary

The results of the testing phase indicate that the implemented modules of the system function correctly and meet the defined functional requirements for the current development phase.

The authentication system, role-based access control, patient management module, device registration module, and dashboard visualization are stable and operational.

The system is ready for demonstration and further enhancement in subsequent development phases.